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Immunostimulating IL-2 analogs

Abstract

Modifications to interleukin-2 alpha receptors are disclosed. Interleukin-2 analogs with increased binding affinity for interleukin-2 beta receptors are disclosed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional of U.S. application Ser. No. 17/915,808 filed Sep. 29, 2022, which is National Stage of International Application No. PCT/KR2021/004028 filed Mar. 31, 2021, claiming priority based on Korean Patent Application No. 10-2020-0039476 filed Mar. 31, 2020.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING
(1) The content of the electronically submitted sequence listing, file name: Q288594_SEQ_LIS_ST26_AS_FILED.XML; size: 595,082 bytes; and date of creation: Jul. 7, 2023, filed herewith, is incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present invention relates to a novel interleukin-2 analog.
BACKGROUND ART
(3) Interleukin-2 is an important immunostimulator with a molecular weight of about 15 kDa, which consists of a total of 133 amino acid residues, and activates various cells of the immune system including T cells and B cells. The high efficacy of interleukin-2 as an immune stimulant can be used for the treatment of various immune-related conditions including cancer and AIDS (Korean Patent Application Publication No. 10-2017-0070091). Currently, interleukin-2 (trademark name: Proleukin) is an FDA-approved drug for the treatment of metastatic renal cell carcinoma and metastatic melanoma. However, due to the severe toxicity associated with high-dose interleukin-2 therapy, the applicable patients are limited. In fact, this therapy is applied to only a small number of eligible patients. The toxicity associated with interleukin-2 includes severe fever, nausea, vomiting, vascular leak, severe hypotension, pulmonary edema, and vascular leak syndrome, which causes liver damage.
(4) The interleukin-2 receptor has three kinds of subunit receptors. The subunit consists of an alpha

chain (IL-2Rα, CD25), a beta chain (IL-2Rβ or CD122), and a gamma chain (IL-2Rγ or CD132). Interleukin-2 can exhibit various functions by binding to receptor subunits of various combinations. A single interleukin-2 alpha receptor is called a low-affinity interleukin-2 receptor, and it is not involved in signaling. A complex of interleukin-2 beta and gamma receptors binds to interleukin-2 with intermediate affinity. A complex of interleukin-2 alpha, beta, and gamma receptors binds to interleukin-2 with high affinity. The complex of interleukin-2 beta and gamma receptors is required for effective signal conversion through kinase activation in multiple signaling pathways. In particular, interleukin-2 beta- and gamma-binding receptors are prominent in CD8+ cells and natural killer (NK) cells. In addition, complexes of high-affinity interleukin-2 alpha, beta, and gamma receptors are usually found in CD4+ T regulatory cells (Treg), and recently they were also found in activated T cells. Since interleukin-2 beta receptors are distributed in CD8+ T cells or natural killer cells (NK cells) and are involved in the immune response in the body, studies have been conducted to develop therapeutic agents by increasing the activity of beta receptors for immune activation.

(5) Meanwhile, despite the potential of interleukin-2 as a therapeutic agent for various immune-related conditions, there are still not many drugs which can reduce their doses while reducing toxicity and side effects, and thus there is an increasing demand for studies on new and improved drugs.

DISCLOSURE

Technical Problem

(6) An object of the present invention is to provide an interleukin-2 analog.

(7) Another object of the present invention is to provide an isolated nucleic acid encoding the interleukin-2 analog; a recombinant expression vector including the nucleic acid; and a transformant including the vector.

(8) Still another object of the present invention is to provide a method for preparing the interleukin-2 analog.

(9) Still another object of the present invention is to provide a method for increasing the binding affinity for interleukin-2 beta receptors, which includes modifying one or more amino acids in native interleukin-2.

Technical Solution

(10) An aspect of the present invention provides a novel interleukin-2 analog (or IL-2 analog). The interleukin-2 analog is an interleukin-2 analog which has an increased binding affinity for interleukin-2 beta receptor compared to native interleukin-2 or aldesleukin (i.e., an interleukin-2 analog). In a specific embodiment, the interleukin-2 analog may include a sequence in which one or more amino acids in native interleukin-2 are modified.

(11) In another specific embodiment, the interleukin-2 analog is characterized in that the interleukin-2 analog includes a sequence in which one or more amino acids corresponding to those at positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2 are modified.

(12) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it has an altered binding affinity for interleukin-2 alpha receptors and an increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 or aldesleukin.

(13) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that one or more amino acids are added to the amino acid corresponding to position 133.

(14) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it is native interleukin-2, in which the amino acid at position 1 is removed and the amino acid at position 125 is substituted with a different amino acid.

(15) The interleukin-2 analog according to any one of the previous specific embodiments is

characterized in that it further includes 1 to 10 amino acid substitutions. The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that one or more amino acids at positions 18, 19, 20, 22, 38, 42, 43, 45, 61, 68, 69, 74, 80, 81, 84, 85, 86, 88, 89, 91, 92, 94, and 96 are further substituted with different amino acids.

(16) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that one or more amino acids at positions 18, 19, 22, 38, 42, 43, 45, 61, 68, 74, 80, 81, 84, 85, 86, 88, 91, 92, 94, and 96 are further substituted with different amino acids.

(17) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it is any one of the following analogs: (a) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 32 are substituted with different amino acids; (b) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 35 are substituted with different amino acids; (c) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 38 are substituted with different amino acids; (d) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 42 are substituted with different amino acids; (e) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 43 are substituted with different amino acids; (f) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 48 are substituted with different amino acids; (g) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 49 are substituted with different amino acids; (h) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 76 are substituted with different amino acids; (i) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 92, 94, and 96 are substituted with different amino acids; (j) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 87 are substituted with different amino acids; (k) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 42 are substituted with different amino acids; (l) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 80 are substituted with different amino acids; (m) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 84 are substituted with different amino acids; (n) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 19, 38, and 42 are substituted with different amino acids; (o) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 12, 38, and 42 are substituted with different amino acids; (p) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 61 are substituted with different amino acids; (q) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 84 are substituted with different amino acids; (r) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at

positions 125, 38, 42, and 88 are substituted with different amino acids; (s) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 89 are substituted with different amino acids; (t) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 91 are substituted with different amino acids; (u) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 94 are substituted with different amino acids; (v) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 126 are substituted with different amino acids; (w) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, and 84 are substituted with different amino acids; (x) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 94, and 96 are substituted with different amino acids; (y) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, and 92 are substituted with different amino acids; (z) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 61, 81, and 92 are substituted with different amino acids; (aa) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, and 92 are substituted with different amino acids; (ab) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, and 92 are substituted with different amino acids, (ac) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, 84, and 92 are substituted with different amino acids; (ad) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 20, 38, 42, 81, and 92 are substituted with different amino acids, (ae) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, and 92 are substituted with different amino acids, (af) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 81, and 92 are substituted with different amino acids; (ag) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 84, and 92 are substituted with different amino acids, (ah) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 88, and 92 are substituted with different amino acids; (ai) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 85, 86, and 92 are substituted with different amino acids, (aj) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, and 92 are substituted with different amino acids; (ak) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 86, and 92 are substituted with different amino acids; (al) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, and 92 are substituted with different amino acids; (am) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 81, and 92 are substituted with different amino acids; (an) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 80, 81, and 92 are substituted with different amino acids; (ao) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 84, and 92 are substituted with different amino acids; (ap) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 45, 80, 85, 86, and 92 are substituted with different amino acids; (aq) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ar) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (as) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (at) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 69, 80, 81, 85, 86, and 92 are substituted with different amino acids; (au) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (av) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (aw) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ax) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 80, 81, and 92 are substituted with different amino acids; (ay) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (az) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 69, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ba) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 84, 85, 86, 91, and 92 are substituted with different amino acids; (bb) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 92, 94, and 96 are substituted with different amino acids; (bc) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 19, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bd) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (be) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bf) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bg) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2

in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bh) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bi) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (bj) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids; (bk) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bl) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bm) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids; (bn) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids; (bo) an interleukin-2 analog in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids in native interleukin-2; and (bp) an interleukin-2 analog in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids in native interleukin-2.
(18) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it includes any one amino acid substitution selected from the group consisting of the following amino acid substitutions: (a) a substitution in which the amino acid at position 12 is substituted with valine or phenylalanine; (b) a substitution in which the amino acid at position 18 is substituted with arginine; (c) a substitution in which the amino acid at position 19 is substituted with tyrosine, valine, phenylalanine, or arginine; (d) a substitution in which the amino acid at position 20 is substituted with valine or phenylalanine; (e) a substitution in which the amino acid at position 22 is substituted with glutamic acid; (f) a substitution in which the amino acid at position 32 is substituted with cysteine; (g) a substitution in which the amino acid at position 35 is substituted with cysteine or glutamic acid; (h) a substitution in which the amino acid at position 38 is substituted with alanine or aspartic acid; (i) a substitution in which the amino acid at position 42 is substituted with lysine, alanine, or tryptophan; (j) a substitution in which the amino acid at position 43 is substituted with cysteine, glutamic acid, or glutamine; (k) a substitution in which the amino acid at position 45 is substituted with alanine; (l) a substitution in which the amino acid at position 48 is substituted with cysteine; (m) a substitution in which the amino acid at position 49 is substituted with cysteine; (n) a substitution in which the amino acid at position 61 is substituted with glutamine, arginine, or aspartic acid; (o) a substitution in which the amino acid at position 68 is substituted with aspartic acid or glutamine; (p) a substitution in which the amino acid at position 69 is substituted with glycine; (q) a substitution in which the amino acid at position 74 is substituted with histidine or alanine; (r) a substitution in which the amino acid at position 76 is substituted with cysteine; (s) a substitution in which the amino acid at position 80 is substituted with phenylalanine, tyrosine, valine, aspartic acid, or tryptophan; (t) a substitution in which the amino acid at position 81 is substituted with aspartic acid, glutamic acid, or asparagine; (u) a substitution in which the amino acid at position 82 is substituted with glycine or valine; (v) a substitution in which the amino acid at position 84 is substituted with glutamic acid, valine, or phenylalanine; (w) a substitution in which the amino acid at position 85 is substituted with valine,

alanine, glycine, tryptophan, tyrosine, threonine, isoleucine, glutamic acid, or phenylalanine; (x) a substitution in which the amino acid at position 86 is substituted with valine, alanine, glycine, or leucine; (y) a substitution in which the amino acid at position 87 is substituted with cysteine; (z) a substitution in which the amino acid at position 88 is substituted with glutamine, valine, or phenylalanine; (aa) a substitution in which the amino acid at position 89 is substituted with phenylalanine; (ab) a substitution in which the amino acid at position 91 is substituted with threonine, phenylalanine, or glutamic acid; (ac) a substitution in which the amino acid at position 92 is substituted with phenylalanine, leucine, tyrosine, or tryptophan; (ad) a substitution in which the amino acid at position 95 is substituted with aspartic acid; (ae) a substitution in which the amino acid at position 96 is substituted with phenylalanine, valine, or isoleucine; and (af) a substitution in which the amino acid at position 126 is substituted with threonine.

(19) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it is selected from the group consisting of SEQ ID NOS: 3 to 106.

(20) Another aspect to implement the present invention provides an isolated nucleic acid encoding the interleukin-2 analog; a recombinant expression vector which includes the nucleic acid; and a transformant which includes the vector.

(21) Still another aspect to implement the present invention provides a method for preparing the interleukin-2 analog.

(22) Still another aspect to implement the present invention provides a method for increasing the binding affinity for interleukin-2 beta receptors, which includes modifying one or more amino acids in native interleukin-2, in which the modification may be modifications in one or more amino acids selected from the group consisting of amino acids corresponding to positions at 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133.

(23) Still another aspect to implement the present invention provides an interleukin-2 analog which includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 3 to 106.

(24) Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence represented by General Formula 1 below:

(25) TABLE-US-00001 [General Formula 1] (General Formula 1, SEQ ID NO: 212) X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-X19-D-L- X22-M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42- X43-F-X45-M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K- P-L-E-X68-V-L-N-L-A-X74-S-K-N-F-H-X80-X81-P-R-X84- X85-X86-S-N-I-N-X91-X92-V-X94-E-X96-K-G-S-E-T-T-F- M-C-E-Y-A-D-E-T-A-T-I-V-E-F-L-N-R-W-I-T-F-S-Q-S-I- I-S-T-L-T wherein in General Formula 1 above, X1 is a deletion; X18 is leucine (L) or arginine (R); X19 is leucine (L) or tyrosine (Y); X22 is glutamic acid (E) or glutamine (Q); X38 is alanine (A), aspartic acid (D), or arginine (R); X42 is alanine (A), phenylalanine (F), lysine (K), or tryptophan (W); X43 is glutamic acid (E), lysine (K), or glutamine (Q); X45 is alanine (A) or tyrosine (Y); X61 is aspartic acid (D), glutamic acid (E), glutamine (Q), or arginine (R); X68 is aspartic acid (D) or glutamic acid (E); X74 is histidine (H) or glutamine (Q); X80 is phenylalanine (F), leucine (L), valine (V), or tyrosine (Y); X81 is aspartic acid (D), glutamic acid (E), or arginine (R); X84 is aspartic acid (D) or glutamic acid (E); X85 is alanine (A), glutamic acid (E), glycine (G), leucine (L), valine (V), tryptophan (W), or tyrosine (Y); X86 is alanine (A), glycine (G), isoleucine (I), or valine (V); X91 is threonine (T) or valine (V); X92 is phenylalanine (F), isoleucine (I), or tyrosine (Y); X94 is phenylalanine (F) or leucine (L); and X96 is phenylalanine (F) or leucine (L).

(26) In a specific embodiment, the interleukin-2 analog is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 15, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 56, 58, 59, 60, 62, 71, 72, 74, 75, 76, 77, 78, 85, 87, 89, 91, 92, 93, 94, 95, 98, 99, 100, 101, 103, 104, 105, and 106.

(27) In another specific embodiment, the interleukin-2 analog is characterized in that in General

Formula 1 above, X43 is lysine (K); X45 is tyrosine (Y); X61 is aspartic acid (D), glutamic acid (E), or glutamine (Q); X68 is glutamic acid (E); X74 is glutamine (Q); X80 is phenylalanine (F) or leucine (L); X85 is leucine (L), valine (V), or tyrosine (Y); X86 is isoleucine (I) or valine (V); and X92 is phenylalanine (F) or isoleucine (I).
(28) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 87, 89, 91, 92, 93, 94, 98, 99, 100, 101, 103, 104, and 105.
(29) The interleukin-2 analog according to any one of the previous specific embodiments is characterized in that it further includes one or more amino acids at the C-terminus thereof.
(30) Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence represented by General Formula 2 below:
(31) TABLE-US-00002 [General Formula 2] (General Formula 2, SEQ ID NO: 213) X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-L-D-L-X22- M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-K-F-Y- M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-P-L-E-X68- V-L-N-L-A-Q-S-K-N-F-H-F-X81-P-R-D-X85-X86-S-N-I-N- V-F-V-L-E-L-K-G-S-E-T-T-F-M-C-E-Y-A-D-E-T-A-T-I-V- E-F-L-N-R-W-I-T-F-S-Q-S-I-I-S-T-L-T wherein in General Formula 2 above, X1 is a deletion, X18 is leucine (L) or arginine (R); X22 is glutamic acid (E) or glutamine (Q); X38 is alanine (A) or arginine (R); X42 is phenylalanine (F) or lysine (K); X61 is aspartic acid (D) or glutamic acid (E); X68 is aspartic acid (D) or glutamic acid (E); X81 is aspartic acid (D) or glutamic acid (E); X85 is leucine (L) or valine (V); and X86 is isoleucine (I) or valine (V).
(32) In a specific embodiment, the interleukin-2 analog is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 22, 42, 53, 87, 105, and 106.
(33) In another specific embodiment, the interleukin-2 analog is characterized in that it further includes one or more amino acids at the C-terminus thereof.
Advantageous Effects
(34) The interleukin-2 analog according to the present invention is an analog which has increased binding affinity for interleukin-2 beta receptors in vivo and can be for various purposes.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1**(A)-FIG. **1**(C) show the results confirming the binding affinity of an interleukin-2 analog for interleukin-2 alpha receptors, in which FIG. **1**(A) represents interleukin-2 analog #86, FIG. **1**(B) represents interleukin-2 analog #104, and FIG. **1**(C) represents interleukin-2 analog #105.
(2) FIG. **2**(A)-FIG. **2**(C) show the results confirming the binding affinity of an interleukin-2 analog for interleukin-2 beta receptors, in which FIG. **2**(A) represents interleukin-2 analog #86, FIG. **2**(B) represents interleukin-2 analog #104, and FIG. **2**(C) represents interleukin-2 analog #105.
DETAILED DESCRIPTION FOR CARRYING OUT THE INVENTION
(3) The details for carrying out the present invention will be described as follows. Meanwhile, respective descriptions and embodiments disclosed in the present invention may also be applied to other descriptions and embodiments. That is, all combinations of various elements disclosed in the present invention fall within the scope of the present invention. Further, the scope of the present invention cannot be considered to be limited by the specific description below.
(4) Over the entire specification of the present invention, the conventional one-letter and three-letter codes for amino acids are used. Additionally, the amino acids mentioned herein are abbreviated according to the nomenclature rules of the IUPAC-IUB as follows:
(5) TABLE-US-00003 alanine A arginine R asparagine N aspartic acid D cysteine C glutamic acid

E glutamine Q glycine G histidine H isoleucine I leucine L lysine K methionine M phenylalanine F proline P serine S threonine T tryptophan W tyrosine Y valine V

(6) Still another aspect of the present invention provides an interleukin-2 analog. The interleukin-2 analog of the present invention is characterized in that its binding affinity for interleukin-2 receptors is altered, and in particular in that it has increased binding affinity for interleukin-2 beta receptors. Specifically, the interleukin-2 analog of the present invention may be one which has increased binding affinity for interleukin-2 beta receptors, and more specifically one which has altered (increased or decreased) binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or known aldesleukin.

(7) As used herein, the term "interleukin-2 (IL-2)" refers to a type of cytokine which transmits signals in the immune system in vivo. The interleukin-2 is generally known as an important immunostimulator with a size of about 15 kDa.

(8) As used herein, the term "interleukin-2 analog" refers to native interleukin-2 in which one or more amino acids in the sequence thereof are modified. Particularly in the present invention, the interleukin-2 analog may be one which has reduced or increased binding affinity for interleukin-2 receptors compared to native interleukin-2, in which amino acid(s) in native interleukin-2 is(are) modified. The interleukin-2 analog of the present invention may be one which is not naturally occurring.

(9) The native interleukin-2 may be a human interleukin-2, and its sequence may be obtained from known databases, etc. Specifically, the native interleukin-2 may have an amino acid sequence of SEQ ID NO: 1, but is not limited thereto.

(10) In the present invention, what is meant by "native interleukin-2 may have an amino acid sequence of SEQ ID NO: 1" is that not only the sequence which is the same as SEQ ID NO: 1, but also sequences which have a homology of 80% or higher, 85% or higher, 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and 99% or higher to SEQ ID NO: 1 belong to the scope of native interleukin-2 of the present invention; and that the corresponding position(s) of amino acid modification is(are) altered on the amino acid sequence of SEQ ID NO: 1 when sequences having a homology of 80% or higher, 85% or higher, 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and 99% or higher are aligned based on SEQ ID NO: 1.

(11) In the present invention, what is meant by "one or more amino acids in the native sequence are altered" may be that a modification selected from the group consisting of substitution, addition, deletion, modification, and a combination thereof has occurred in at least one amino acid in the native sequence.

(12) Specifically, the interleukin-2 analog of the present invention may include a sequence in which one or more amino acids corresponding to positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2 are modified. Specifically, the interleukin-2 analog of the present invention may be native interleukin-2 in which the amino acid at position 1 is removed and the amino acid at position 125 is substituted with a different amino acid; and which further includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more amino acid substitutions. Although not limited thereto, the amino acid at position 125 (i.e., cysteine) may be substituted with serine, and the amino acid(s) at which a further substitution occurs may be amino acids corresponding to positions 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 126, and 133.

(13) Additionally, interleukin-2 analogs which include substitution, addition, deletion, modification, etc. of amino acid residues in addition to the positions for modification above to the extent that can be performed for the stability and increase of half-life of a peptide known in the art are also included within the scope of the present invention.

(14) As used herein, the term “aldesleukin” or “interleukin-2 analog (aldesleukin)”, which is a commercially available interleukin-2 analog, may be aldesleukin (trademark name: Proleukin®), and specifically may be one which has the amino acid sequence of SEQ ID NO: 2. In the present invention, these terms are used interchangeably with “interleukin-2 analog 1”. The interleukin analog according to the present invention may have altered binding affinity for interleukin-2 alpha receptors and/or increased binding affinity for interleukin-2 beta receptors compared to the interleukin-2 analog 1.
(15) Although interleukin-2 alpha receptors are not known to be involved in the signaling system of interleukin-2, they increase the binding affinity of interleukin-2 for other interleukin-2 receptors (beta or gamma receptors) by 10 to 100 times and are expressed in CD4.sup.+ regulatory T cells, etc.
(16) Since interleukin-2 beta receptors are mainly distributed in CD8.sup.+ T cells or natural killer cells (NK cells) and have an important role of activating immune responses and macrophages, it is expected that tumor cell death and activation of the body's immune responses can be promoted through the activation of interleukin-2 beta receptors.
(17) Accordingly, the interleukin-2 analog of the present invention which has increased binding affinity for interleukin-2 beta receptors can have a therapeutic effect where the suppression and death of tumors is increased while side effects are reduced.
(18) In the present invention, the interleukin-2 analog may include a sequence of native interleukin-2 in which the amino acid at position 1 is removed and the amino acid at position 125 is substituted with a different amino acid, and which further includes 1 to 10 amino acid modifications. For example, the interleukin-2 analog may include a sequence of native interleukin-2 in which the amino acid at position 125 is substituted with serine and one or more amino acids at positions 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, and 126 are substituted with different amino acids and/or one or more amino acids are added on the amino acid at position 133, but the sequence is not limited thereto, and any interleukin-2 analog which has altered binding affinity for interleukin-2 alpha receptors and increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 and/or aldesleukin is included without limitation.
(19) In an embodiment, the interleukin-2 analog may be one in which one or more amino acids are added to the amino acid corresponding to position 133, but the interleukin-2 analog is not limited thereto. For the purpose of the present invention, the amino acids to be added are not limited with regard to the type or length thereof as long as the interleukin-2 analog has altered binding affinity for interleukin-2 alpha receptors and increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 and/or aldesleukin, and amino acids which are not naturally occurring and amino acids with a chemical modification can also be added in addition to natural amino acids.
(20) In another embodiment, the interleukin-2 analog may be native interleukin-2 in which the amino acid at position 1 is removed; the amino acid at position 125 is substituted with a different amino acid; and 1, 2, 3, 4, 5, 6, 7, 8, 9, or more amino acids among the amino acids at positions 18, 19, 20, 22, 38, 42, 43, 45, 61, 68, 69, 74, 80, 81, 84, 85, 86, 88, 89, 91, 92, 94, and 96 are substituted with different amino acids, but the interleukin-2 analog is not limited thereto.
(21) In still another embodiment, the interleukin-2 analog may be native interleukin-2 in which the amino acid at position 1 is removed; the amino acid at position 125 is substituted with a different amino acid; and 1, 2, 3, 4, 5, 6, 7, 8, 9, or more amino acids among the amino acids at positions 18, 19, 22, 38, 42, 43, 45, 61, 68, 74, 80, 81, 84, 85, 86, 88, 91, 92, 94, and 96 are substituted with different amino acids, but the interleukin-2 analog is not limited thereto.
(22) In still another embodiment, the interleukin-2 analog may be any one selected from the group consisting of the following analogs: (a) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at

positions 125 and 32 are substituted with different amino acids; (b) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 35 are substituted with different amino acids; (c) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 38 are substituted with different amino acids; (d) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 42 are substituted with different amino acids; (e) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 43 are substituted with different amino acids; (f) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 48 are substituted with different amino acids; (g) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 49 are substituted with different amino acids; (h) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 76 are substituted with different amino acids; (i) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 92, 94, and 96 are substituted with different amino acids; (j) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125 and 87 are substituted with different amino acids; (k) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 42 are substituted with different amino acids; (l) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 80 are substituted with different amino acids; (m) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 84 are substituted with different amino acids; (n) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 19, 38, and 42 are substituted with different amino acids; (o) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 12, 38, and 42 are substituted with different amino acids; (p) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 61 are substituted with different amino acids; (q) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 84 are substituted with different amino acids; (r) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 88 are substituted with different amino acids; (s) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 89 are substituted with different amino acids; (t) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 91 are substituted with different amino acids; (u) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 94 are substituted with different amino acids; (v) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 126 are substituted with different amino acids; (w) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, and 84 are substituted with different amino acids; (x) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 94, and 96 are substituted with different amino acids; (y) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, and 92 are substituted with different amino acids; (z) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 61, 81, and 92 are substituted with different amino acids; (aa) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, and 92 are substituted with different amino acids; (ab) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, and 92 are substituted with different amino acids; (ac) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, 84, and 92 are substituted with different amino acids, (ad) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 20, 38, 42, 81, and 92 are substituted with different amino acids; (ae) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, and 92 are substituted with different amino acids, (af) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 81, and 92 are substituted with different amino acids, (ag) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 84, and 92 are substituted with different amino acids; (ah) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 88, and 92 are substituted with different amino acids, (ai) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 85, 86, and 92 are substituted with different amino acids, (aj) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, and 92 are substituted with different amino acids; (ak) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 86, and 92 are substituted with different amino acids, (al) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, and 92 are substituted with different amino acids; (am) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 81, and 92 are substituted with different amino acids; (an) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 80, 81, and 92 are substituted with different amino acids; (ao) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 84, and 92 are substituted with different amino acids; (ap) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 45, 80, 85, 86, and 92 are substituted with different amino acids; (aq) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ar) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (as) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (at) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 69, 80, 81, 85, 86, and 92 are substituted with different amino acids; (au) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (av) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (aw) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ax) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 80, 81, and 92 are substituted with different amino acids; (ay) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (az) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 69, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ba) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 84, 85, 86, 91, and 92 are substituted with different amino acids; (bb) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 92, 94, and 96 are substituted with different amino acids; (bc) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 19, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bd) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (be) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bf) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bg) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bh) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bi) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (bj) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids; (bk) an interleukin-2 analog, wherein the interleukin-

2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bl) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bm) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids; (bn) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids; (bo) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids; and (bp) an interleukin-2 analog, wherein the interleukin-2 analog is native interleukin-2 in which the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids.
(23) In particular, the amino acid substitutions included in the interleukin-2 analog may be any one or more selected from the group consisting of the following amino acid substitutions: (a) a substitution in which the amino acid at position 12 is substituted with valine or phenylalanine; (b) a substitution in which the amino acid at position 18 is substituted with arginine; (c) a substitution in which the amino acid at position 19 is substituted with tyrosine, valine, phenylalanine, or arginine; (d) a substitution in which the amino acid at position 20 is substituted with valine or phenylalanine; (e) a substitution in which the amino acid at position 22 is substituted with glutamic acid; (f) a substitution in which the amino acid at position 32 is substituted with cysteine; (g) a substitution in which the amino acid at position 35 is substituted with cysteine or glutamic acid; (h) a substitution in which the amino acid at position 38 is substituted with alanine or aspartic acid; (i) a substitution in which the amino acid at position 42 is substituted with lysine, alanine, or tryptophan; (j) a substitution in which the amino acid at position 43 is substituted with cysteine, glutamic acid, or glutamine; (k) a substitution in which the amino acid at position 45 is substituted with alanine; (l) a substitution in which the amino acid at position 48 is substituted with cysteine; (m) a substitution in which the amino acid at position 49 is substituted with cysteine; (n) a substitution in which the amino acid at position 61 is substituted with glutamine, arginine, or aspartic acid; (o) a substitution in which the amino acid at position 68 is substituted with aspartic acid or glutamine; (p) a substitution in which the amino acid at position 69 is substituted with glycine; (q) a substitution in which the amino acid at position 74 is substituted with histidine or alanine; (r) a substitution in which the amino acid at position 76 is substituted with cysteine; (s) a substitution in which the amino acid at position 80 is substituted with phenylalanine, tyrosine, valine, aspartic acid, or tryptophan; (t) a substitution in which the amino acid at position 81 is substituted with aspartic acid, glutamic acid, or asparagine; (u) a substitution in which the amino acid at position 82 is substituted with glycine or valine; (v) a substitution in which the amino acid at position 84 is substituted with glutamic acid, valine, or phenylalanine; (w) a substitution in which the amino acid at position 85 is substituted with valine, alanine, glycine, tryptophan, tyrosine, threonine, isoleucine, glutamic acid, or phenylalanine; (x) a substitution in which the amino acid at position 86 is substituted with valine, alanine, glycine, or leucine; (y) a substitution in which the amino acid at position 87 is substituted with cysteine; (z) a substitution in which the amino acid at position 88 is substituted with glutamine, valine, or phenylalanine; (aa) a substitution in which the amino acid at position 89 is substituted with phenylalanine; (ab) a substitution in which the amino acid at position 91 is substituted with threonine, phenylalanine, or glutamic acid; (ac) a substitution in which the amino acid at position 92 is substituted with phenylalanine, leucine, tyrosine, or tryptophan; (ad) a substitution in which the amino acid at position 95 is substituted with aspartic acid; (ae) a substitution in which the amino acid at position 96 is substituted with phenylalanine,

valine, or isoleucine; and (af) a substitution in which the amino acid at position 126 is substituted with threonine.
(24) As used herein, the term "corresponding to" refers to an amino acid residue at a position listed in a peptide, or an amino acid residue which is similar, identical, or homologous to a residue listed in a peptide. Confirmation of the amino acid at the corresponding position may be determining a specific amino acid in a sequence that refers to a specific sequence.
(25) In an embodiment, each amino acid residue in the amino acid sequence can be numbered by aligning any amino acid sequence with SEQ ID NO: 1, and based on the same, referring to the numerical position of the amino acid residue corresponding to the amino acid residue of SEQ ID NO: 1.
(26) As such an alignment, for example, the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970, *J. Mol. Biol.* 48:443-453), the Needle program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000), (*Trends Genet.* 16:276-277), etc. may be used, but the available programs are not limited thereto, and a sequence alignment program known in the art, a pairwise sequence comparison algorithm, etc. may be used appropriately.
(27) In the present invention, even if expressed as a specific position of an amino acid in a peptide, such expression may refer to a corresponding position in a reference sequence.
(28) In another embodiment, the interleukin-2 analog may include, consist of or essentially consist of an amino acid sequence which is selected from the group consisting of SEQ ID NOS: 3 to 106, but the interleukin-2 analog is not limited thereto.
(29) Additionally, even if the interleukin-2 analog is expressed as "an interleukin-2 analog consisting of a particular SEQ ID NO" in the present invention, it does not exclude a mutation that may occur by the addition of a meaningless sequence upstream or downstream of the amino acid sequence of the corresponding SEQ ID NO, or a mutation that may occur naturally, or a silent mutation thereof, as long as the interleukin-2 analog has an activity identical or equivalent to the interleukin-2 analog consisting of the amino acid sequence of the corresponding SEQ ID NO, and even if the sequence addition or mutation is present, the interleukin-2 analog apparently belongs to the scope of the present invention.
(30) The interleukin-2 analog of the present invention may include an amino acid sequence which has a homology of 40% or higher, 45% or higher, 50% or higher, 55% or higher, 60% or higher, 65% or higher, 70% or higher, 75% or higher, 80% or higher, 81% or higher, 82% or higher, 83% or higher, 84% or higher, 85% or higher, 86% or higher, 87% or higher, 88% or higher, 89% or higher, 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, or 99% or higher to the amino acid sequences of SEQ ID NOS: 3 to 106, but the interleukin-2 analog is not limited thereto.
(31) In the present invention, the terms "homology" and "identity" refer to a degree of relatedness between two given amino acid sequences or nucleotide sequences and may be expressed as a percentage.
(32) Sequence homology or identity of a conserved polynucleotide or polypeptide may be determined by a standard alignment algorithm, and default gap penalties established by a program to be used may be used in combination. Substantially, homologous or identical sequences may generally hybridize with all or part of the sequences under moderately or highly stringent conditions. It is apparent that hybridization also includes hybridization of a polynucleotide with a polynucleotide, which includes a general codon or a codon where codon degeneracy is considered.
(33) The terms homology and identity can frequently be used interchangeably.
(34) Whether any two nucleotide or peptide sequences have homology, similarity, or identity may be determined by, for example, a known computer algorithm (e.g., the "FASTA" program) using default parameters as in Pearson et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:2444). Alternatively, they may be determined using the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970,

J. Mol. Biol. 48:443-453) as performed in the Needleman program of the EMBOSS package (EMBOSS: The European Molecular Biology Open Software Suite, Rice et al., 2000, *Trends Genet.* 16:276-277) (version 5.0.0 or later) (including GCG program package (Devereux, J. et al., *Nucleic Acids Research* 12:387 (1984)), BLASTP, BLASTN, FASTA (Atschul, S. F. et al., *J Molec Biol* 215:403 (1990); *Guide to Huge Computers*, Martin J. Bishop, ed., Academic Press, San Diego, 1994, and CARILLO et al. (1988) *SIAM J Applied Math* 48:1073). For example, homology, similarity, or identity may be determined using BLAST or ClustalW of the National Center for Biotechnology Information.

(35) Homology, similarity, or identity of nucleotide or peptide sequences may be determined by comparing sequence information using the GAP computer program (e.g., Needleman et al. (1970), *J Mol Biol* 48:443) as disclosed in Smith and Waterman, *Adv. Appl. Math* (1981) 2:482. Briefly, the GAP program defines homology, similarity, or identity as the number of similar aligned symbols (i.e., nucleotides or amino acids), divided by the total number of symbols in the shorter of the two sequences. The default parameters for the GAP program may include: (1) a unary comparison matrix (containing a value of 1 for identities and 0 for non-identities) and the weighted comparison matrix (or EDNAFULL (EMBOSS version of NCBI NUC4.4) substitution matrix) of Gribskov et al. (1986) *Nucl. Acids Res.* 14:6745, as disclosed by Schwartz and Dayhoff, eds., *Atlas Of Protein Sequence And Structure*, National Biomedical Research Foundation, pp. 353-358 (1979); (2) a penalty of 3.0 for each gap and an additional 0.10 penalty for each symbol in each gap (or gap open penalty 10, gap extension penalty 0.5); and (3) no penalty for end gaps. Therefore, the terms “homology” and “identity”, as used herein, represent relevance between sequences.

(36) The above may be applied to other embodiments or other aspects of the present invention, but is not limited thereto.

(37) The interleukin-2 analog of the present invention may be used as a novel interleukin-2 substitute that alters its in vitro activity by weakening or increasing the binding affinity of the interleukin-2 analog for interleukin-2 alpha and/or beta receptors. In particular, the interleukin-2 analog of the present invention can be used as an effective therapeutic agent due to its activities for the two types of receptors because it not only has an increased binding affinity for beta receptors but also has altered (i.e., increased or decreased) binding affinity for alpha receptors.

(38) In the present invention, such modification for preparing analogs of interleukin-2 includes all of the modifications using L-type or D-type amino acids and/or non-natural amino acids; and/or a modification of native sequence, for example, a modification of a side chain functional group, an intramolecular covalent bonding (e.g., a ring formation between side chains), methylation, acylation, ubiquitination, phosphorylation, aminohexanation, biotinylation, etc.

(39) Additionally, the modification includes all of those where one or more amino acids are added to the amino and/or carboxy terminus of native interleukin-2.

(40) As the amino acids to be substituted or added, not only the 20 amino acids commonly observed in human proteins, but also atypical amino acids or those which are not naturally occurring can be used. Commercial sources of atypical amino acids include Sigma-Aldrich, Chem Pep Inc., and Genzyme Pharmaceuticals. The peptides including these amino acids and typical peptide sequences may be synthesized and purchased from commercial suppliers, e.g., American Peptide Company, Bachem (USA), or Anygen (Korea).

(41) Amino acid derivatives may be obtained in the same manner, and as one such example, 4-imidazoacetic acid, etc. may be used.

(42) Additionally, the interleukin-2 analog according to the present invention may be in a modified form where the N-terminus and/or C-terminus, etc. of the interleukin-2 is chemically modified or protected by organic groups, or amino acids may be added to the terminus of the peptide, etc. for its protection from proteases in vivo while increasing its stability.

(43) In particular, in the case of a chemically synthesized peptide, its N- and C-termini are electrically charged, and thus the N-terminus of the peptide may be acetylated and/or C-terminus of

the peptide may be amidated, but the peptide is not particularly limited thereto.
(44) Additionally, since the interleukin-2 analog according to the present invention is in a peptide form, it may include all of those in the form of the peptide itself, a salt thereof (e.g., a pharmaceutically acceptable salt of the peptide), or a solvate thereof. Additionally, the peptide may be in any pharmaceutically acceptable form.
(45) The kind of the salt is not particularly limited. However, it is desirable that the salt be in a safe and effective form for a subject (e.g., a mammal), but the salt type is not particularly limited thereto.
(46) The term “pharmaceutically acceptable” refers to a material which can be effectively used for a desired purpose without causing excessive toxicity, irritation, allergic reactions, etc. within the scope of medical judgment.
(47) As used herein, the term “pharmaceutically acceptable salt” includes salts which are derived from pharmaceutically acceptable inorganic acids, organic acids, or bases. Examples of suitable acids include hydrochloric acid, bromic acid, sulfuric acid, nitric acid, perchloric acid, fumaric acid, maleic acid, phosphoric acid, glycolic acid, lactic acid, salicylic acid, succinic acid, toluene-p-sulfonic acid, tartaric acid, acetic acid, citric acid, methanesulfonic acid, formic acid, benzoic acid, malonic acid, naphthalene-2-sulfonic acid, benzenesulfonic acid, etc. Salts derived from suitable bases may include alkali metals (e.g., sodium, potassium, etc.), alkaline earth metals (e.g., magnesium, etc.), ammonium, etc.
(48) Additionally, as used herein, the term “solvate” refers to a complex formed between the peptide according to the present invention or a salt thereof and a solvent molecule.
(49) In the present invention, the binding affinity of any interleukin-2 analog for native interleukin-2 receptors can be measured using various known techniques, which are methods for measuring the affinity for the receptors. For example, surface plasmon resonance (SPR) may be used, but the measurement method is not limited thereto.
(50) More specifically, the interleukin-2 analog of the present invention may have reduced or increased binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or aldesleukin.
(51) Specifically, the interleukin-2 analog of the present invention may have binding affinity for interleukin-2 alpha receptors of about 0.001-fold or greater, about 0.005-fold or greater, about 0.01-fold or greater, about 0.05-fold or greater, about 0.1-fold or greater, about 0.3-fold or greater, about 0.5-fold or greater, about 0.7-fold or greater, about 0.9-fold or greater, about 1.1-fold or greater, about 1.3-fold or greater, about 1.5-fold or greater, or about 1.7-fold or greater compared to the binding affinity of native interleukin-2 or aldesleukin for interleukin-2 alpha receptors, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is a change in the binding affinity compared to that of native interleukin-2 or aldesleukin.
(52) Alternatively, based on the binding affinity of aldesleukin for interleukin-2 alpha receptors (set at 100%), the interleukin-2 analog of the present invention may have no binding affinity for interleukin-2 alpha receptors or have binding affinity for interleukin-2 alpha receptors of about 1% or greater, about 5% or greater, about 7% or greater, about 10% or greater, about 15% or greater, about 20% or greater, about 30% or greater, about 50% or greater, about 70% or greater, about 90% or greater, about 100% or greater, about 150% or greater, or about 200% or greater, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is a change in the binding affinity compared to that of native interleukin-2 or aldesleukin.
(53) Additionally, the interleukin-2 analog of the present invention may specifically have binding affinity for interleukin-2 beta receptors of about 0.1-fold or greater, about 0.3-fold or greater, about 0.5-fold or greater, about 0.7-fold or greater, about 1.0-fold or greater, about 10-fold or greater, about 20-fold or greater, about 30-fold or greater, about 40-fold or greater, about 50-fold or greater,

about 60-fold or greater, about 70-fold or greater, about 80-fold or greater, about 90-fold or greater, or about 100-fold or greater compared to the binding affinity of native interleukin-2 or aldesleukin for interleukin-2 beta receptors, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is a change or increase in the binding affinity compared to that of native interleukin-2 or aldesleukin.

(54) Alternatively, based on the binding affinity of aldesleukin for interleukin-2 beta receptors (set at 100%), the interleukin-2 analog of the present invention may have binding affinity for interleukin-2 beta receptors of about 5% or greater, about 9% or greater, about 10% or greater, about 20% or greater, about 30% or greater, about 50% or greater, about 100% or greater, about 200% or greater, about 500% or greater, about 700% or greater, about 1,000% or greater, about 1,500% or greater, about 3,000% or greater, about 5,000% or greater, about 7,000% or greater, about 10,000% or greater, about 12,000% or greater, about 15,000% or greater, about 20,000% or greater, or about 25,000%, but the numerical value of the binding affinity is not limited, and the value will belong to the scope of the present invention as long as there is an increase in the binding affinity compared to that of native interleukin-2 or aldesleukin.

(55) As used herein, the term “about” refers to a range which includes all of ±0.5, ±0.4, ±0.3, ±0.2, ±0.1, etc. and includes all of the values that are equivalent or similar to those following the values, but the range is not limited thereto.

(56) The interleukin-2 analog of the present invention is characterized in that it has altered binding affinity for interleukin-2 alpha receptors and increased binding affinity for interleukin-2 beta receptors compared to native interleukin-2 or aldesleukin.

(57) In a specific embodiment of the present invention, for the preparation of the interleukin-2 analog of the present invention, an interleukin-2 analog was prepared into which a modification was introduced based on native interleukin-2 (SEQ ID NO: 1). The interleukin-2 analog prepared in the present invention may be one which includes any one amino acid sequence among SEQ ID NOS: 3 to 106, or may be one which is encoded by any one nucleotide sequence among SEQ ID NOS: 108 to 211.

(58) Still another aspect to implement the present invention provides a nucleic acid (polynucleotide) encoding the interleukin-2 analog, a recombinant expression vector including the nucleic acid, and a transformant which includes the nucleic acid or recombinant expression vector.

(59) The nucleic acid encoding the interleukin-2 analog of the present invention may be one which is modified so that a modification (deletion, substitution, and/or addition of an amino acid) can be introduced into an amino acid at a particular position in a native interleukin-2 of SEQ ID NO: 1, and specifically, the interleukin-2 analog of the present invention may include a nucleotide sequence encoding any one amino acid sequence among SEQ ID NOS: 3 to 106. For example, the nucleic acid of the present invention may have or include a nucleotide sequence of any one among SEQ ID NOS: 108 to 211.

(60) The nucleotide sequence of the present invention may be modified variously in the coding region within a range not altering the amino acid sequence of the interleukin-2 analog of the present invention, considering codon degeneracy or the codons preferred in the organism where the nucleic acid of the present invention is to be expressed. Specifically, the nucleic acid of the present invention may have or include a nucleotide sequence which has a homology or identity of 70% or higher, 75% or higher, 80% or higher, 85% or higher, 90% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and less than 100% to any one of the sequences of SEQ ID NOS: 108 to 211; or may consist of or essentially consist of a nucleotide sequence which has a homology or identity of 70% or higher, 75% or higher, 80% or higher, 85% or higher, 90% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, and less than 100% to any one of the sequences of SEQ ID NOS: 108 to 211, but the nucleic acid is not limited thereto.

(61) Additionally, the nucleic acid of the present invention can include, without limitation, a probe which can be prepared from a known gene sequence (e.g., a sequence that can hybridize with a

sequence complementary to all or part of the nucleic acid sequence of the present invention under stringent conditions). The "stringent conditions" refer to conditions that enable specific hybridization between polynucleotides. Such conditions are described in detail in the literature (see J. Sambrook et al., *Molecular Cloning, A Laboratory Manual*, 2nd Edition, Cold Spring Harbor Laboratory press, Cold Spring Harbor, New York, 1989; F. M. Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley & Sons, Inc., New York, 9.50-9.51, 11.7-11.8).
(62) Hybridization requires that two nucleic acids have complementary sequences, although mismatches between bases may be possible depending on hybridization stringency. The term "complementary" is used to describe the relationship between nucleotide bases that can hybridize to each other. For example, with respect to DNA, adenine is complementary to thymine, and cytosine is complementary to guanine. Accordingly, the nucleic acid of the present invention can include isolated nucleic acid fragments complementary to the entire sequence as well as to substantially similar nucleic acid sequences.
(63) The appropriate stringency for hybridizing polynucleotides depends on the length of the polynucleotides and the degree of complementarity, and variables are well known in the art (e.g., Sambrook et al., supra).
(64) The homology and identity are as described above.
(65) The recombination vector according to the present invention may be constructed as a vector for typical cloning or a vector for expression, and may be constructed as a vector for use of eukaryotic or prokaryotic cells as a host cell.
(66) As used herein, the term "vector", which is a recombination vector capable of expressing a target protein in an appropriate host cell, refers to a nucleic acid construct that includes essential control elements operably linked to enable the expression of the nucleic acid insert. In the present invention, it is possible to prepare a recombination vector which includes a nucleic acid encoding an interleukin-2 analog, and the interleukin-2 analog of the present invention can be obtained by transforming or transfecting a host cell with the recombination vector.
(67) As used herein, the term "transformation" refers to a phenomenon in which DNA is introduced into a host cell to allow DNA to be replicated as a factor of a chromosome or by completion of chromosome integration, and external DNA is introduced into cells to artificially cause genetic changes.
(68) The host suitable for the present invention is not particularly limited as long as it enables the expression of the nucleic acid of the present invention. Specific examples of the host that can be used in the present invention include bacteria of the genus *Escherichia* (e.g., *E. coli*); bacteria of the genus *Bacillus* (e.g., *Bacillus subtilis*); bacteria of the genus *Pseudomonas* (e.g., *Pseudomonas putida*); yeasts (e.g., *Pichia pastoris, Saccharomyces cerevisiae*, and *Schizosaccharomyces pombe*); insect cells (e.g., *Spodoptera frugiperda* (SF9)); and animal cells (e.g., CHO, COS, BSC, etc.).
(69) Still another aspect to implement the present invention provides a method for preparing an interleukin-2 analog which includes one or more modifications.
(70) Specifically, the method may include introducing a modification into one or more amino acids selected from the group consisting of amino acids corresponding to those at positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2.
(71) More specifically, the method may be: (a) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 32 are substituted with different amino acids; (b) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 35 are substituted with different amino acids; (c) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 38 are substituted with different amino acids; (d) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 42 are substituted with different amino acids; (e) a method, wherein in native interleukin-2, the

amino acid at position 1 is removed and the amino acids at positions 125 and 43 are substituted with different amino acids; (f) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 48 are substituted with different amino acids; (g) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 49 are substituted with different amino acids; (h) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 76 are substituted with different amino acids; (i) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 92, 94, and 96 are substituted with different amino acids; (j) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 87 are substituted with different amino acids; (k) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 42 are substituted with different amino acids; (l) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 80 are substituted with different amino acids; (m) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 84 are substituted with different amino acids; (n) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 19, 38, and 42 are substituted with different amino acids; (o) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 12, 38, and 42 are substituted with different amino acids; (p) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 61 are substituted with different amino acids; (q) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 84 are substituted with different amino acids; (r) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 88 are substituted with different amino acids; (s) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 89 are substituted with different amino acids; (t) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 91 are substituted with different amino acids; (u) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 94 are substituted with different amino acids; (v) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 126 are substituted with different amino acids; (w) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, and 84 are substituted with different amino acids; (x) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 94, and 96 are substituted with different amino acids; (y) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, and 92 are substituted with different amino acids; (z) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 61, 81, and 92 are substituted with different amino acids; (aa) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, and 92 are substituted with different amino acids; (ab) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, and 92 are substituted with different amino acids; (ac) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, 84, and 92 are substituted with different amino acids; (ad) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 20, 38, 42, 81, and 92 are substituted with different amino acids; (ae) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, and 92 are substituted with different amino acids; (af) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 81, and 92 are substituted with different amino acids; (ag) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 84, and 92 are substituted with different amino acids; (ah) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 88, and 92 are substituted with different amino acids; (ai) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 85, 86, and 92 are substituted with different amino acids; (aj) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, and 92 are substituted with different amino acids; (ak) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 86, and 92 are substituted with different amino acids; (al) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, and 92 are substituted with different amino acids; (am) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 81, and 92 are substituted with different amino acids; (an) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 80, 81, and 92 are substituted with different amino acids; (ao) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 84, and 92 are substituted with different amino acids; (ap) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 45, 80, 85, 86, and 92 are substituted with different amino acids; (aq) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ar) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (as) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (at) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 80, 81, 85, 86, and 92 are substituted with different amino acids; (au) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (av) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (aw) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ax) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 80, 81, and 92 are substituted with different amino acids; (ay) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (az) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ba) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 84, 85, 86, 91, and 92 are substituted with different amino acids; (bb) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 92, 94, and 96 are substituted with different amino acids; (bc) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 19, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bd) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (be) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bf) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bg) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bh) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bi) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (bj) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids; (bk) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bl) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bm) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids; or (bn) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids, (bo) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids; or (bp) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids; but the method is not limited thereto.

(72) The interleukin-2 analog and modification are the same as above.

(73) In another embodiment of a method for preparing the interleukin-2 analog of the present invention, the method for preparing the interleukin-2 analog may include (a) culturing a transformant which includes a nucleic acid encoding the interleukin-2 analog and expressing the interleukin-2 analog; and (b) isolating and purifying the expressed interleukin-2 analog, but the method is not limited to any particular method, and any method known in the art may be used as long as the interleukin-2 analog can be prepared by the same.

(74) In the present invention, the nucleic acid encoding the interleukin-2 analog may include or (essentially) consist of any one nucleotide sequence among SEQ ID NOS: 108 to 211.

(75) The medium used for culturing a transformant in the present invention must meet the requirements for culturing host cells in an appropriate manner. The carbon sources that can be included in the medium for the growth of host cells can be appropriately selected as a decision by those skilled in the art according to the type of transformants being produced, and appropriate culture conditions can be adopted to control the time and amount of culture.

(76) Sugar sources that can be used may include sugars and carbohydrates (e.g., glucose, saccharose, lactose, fructose, maltose, starch, and cellulose); oils and fats (e.g., soybean oil, sunflower oil, castor oil, coconut oil, etc.); fatty acids (e.g., palmitic acid, stearic acid, and linoleic acid); alcohols (e.g., glycerol and ethanol); and organic acids (e.g., acetic acid). These materials can be used individually or as a mixture.

(77) Nitrogen sources that can be used may include peptone, yeast extract, gravy, malt extract, corn steep liquor, soybean meal, and urea, or inorganic compounds (e.g., ammonium sulfate, ammonium chloride, ammonium phosphate, ammonium carbonate, and ammonium nitrate). Nitrogen sources can also be used individually or as a mixture.

(78) Phosphorous sources that can be used may include potassium dihydrogen phosphate or

dipotassium hydrogen phosphate, or corresponding sodium-containing salts thereof. In addition, the culture medium may contain a metal salt (e.g., magnesium sulfate and iron sulfate) required for growth.
(79) Finally, in addition to these materials, essential growth materials (e.g., amino acids and vitamins) may be used. In addition, suitable precursors for culture media may be used. The above-mentioned raw materials can be added in a batch or continuous mode in a manner appropriate to the culture during the cultivation. Basic compounds (e.g., sodium hydroxide, potassium hydroxide, and ammonia) or acidic compounds (e.g., phosphoric acid and sulfuric acid) can be used in an appropriate manner to adjust the pH of the culture. In addition, antifoaming agents (e.g., fatty acid polyglycol esters) may be used to inhibit bubble generation. In order to maintain aerobic conditions, oxygen or oxygen-containing gas (e.g., air) is injected into the culture.
(80) Culturing of the transformant according to the present invention is usually performed at a temperature of 20° C. to 45° C., specifically 25° C. to 40° C. In addition, the culture is continued until the maximum amount of the desired interleukin-2 analog is obtained, and for this purpose, the culture can usually last for 10 to 160 hours.
(81) As described above, if the appropriate culture conditions are established depending on the host cell, the transformant according to the present invention will produce an interleukin-2 analog, and depending on the composition of the vector and the characteristics of the host cell, the interleukin-2 analog produced can be secreted into the cytoplasm of the host cell, into the periplasmic space, or extracellularly.
(82) Proteins expressed in the host cell or outside thereof can be purified in a conventional manner. Examples of purification methods include salting out (e.g.: ammonium sulfate precipitation, sodium phosphate precipitation, etc.), solvent precipitation (e.g., protein fractionation precipitation using acetone, ethanol, etc.), dialysis, gel filtration, ion exchange, chromatography (e.g., reverse-phase column chromatography), ultrafiltration, etc. and can be used alone or in combination.
(83) In a specific embodiment of the present invention, a method for preparing an interleukin-2 analog may include: (a) expressing the interleukin-2 analog; and (b) isolating the expressed interleukin-2 analog.
(84) In a specific embodiment of the present invention, the following steps may be further included to isolate and purify the interleukin-2 analog expressed in the form of an inclusion body from a transformant: (b-1) collecting and disrupting a transformant from the culture medium of the step (a) above; (b-2) recovering and refolding an interleukin-2 analog expressed in a disrupted cell lysate; and (b-3) purifying the refolded interleukin-2 analog by size-exclusion chromatography.
(85) Still another aspect to implement the present invention provides a method for preparing the interleukin-2 analog by way of a peptide synthesis method. Since the sequences of the interleukin-2 analogs of the present invention are already provided, the synthesis of peptides can be performed using a known peptide synthesis method.
(86) The interleukin-2 analog and modification are as described above.
(87) Still another aspect to implement the present invention provides a method for increasing the binding affinity for interleukin-2 beta receptors, which includes modifying one or more amino acids in native interleukin-2.
(88) The method for increasing the binding affinity for interleukin-2 beta receptors according to the present invention may be one which not only increases the binding affinity for interleukin-2 beta receptors but also alters the binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or aldesleukin.
(89) Specifically, the method may include a step of introducing a modification into one or more amino acids corresponding to those at positions 1, 12, 18, 19, 20, 22, 32, 35, 38, 42, 43, 45, 48, 49, 61, 68, 69, 74, 76, 80, 81, 82, 84, 85, 86, 87, 88, 89, 91, 92, 94, 95, 96, 125, 126, and 133 in native interleukin-2.
(90) More specifically, the method may be: (a) a method, wherein in native interleukin-2, the amino

acid at position 1 is removed and the amino acids at positions 125 and 32 are substituted with different amino acids; (b) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 35 are substituted with different amino acids; (c) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 38 are substituted with different amino acids; (d) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 42 are substituted with different amino acids; (e) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 43 are substituted with different amino acids; (f) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 48 are substituted with different amino acids; (g) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 49 are substituted with different amino acids; (h) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 76 are substituted with different amino acids; (i) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 92, 94, and 96 are substituted with different amino acids; (j) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125 and 87 are substituted with different amino acids; (k) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 42 are substituted with different amino acids; (l) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 80 are substituted with different amino acids; (m) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, and 84 are substituted with different amino acids; (n) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 19, 38, and 42 are substituted with different amino acids; (o) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 12, 38, and 42 are substituted with different amino acids; (p) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 61 are substituted with different amino acids; (q) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 84 are substituted with different amino acids; (r) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 88 are substituted with different amino acids; (s) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 89 are substituted with different amino acids; (t) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 91 are substituted with different amino acids; (u) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 94 are substituted with different amino acids; (v) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, and 126 are substituted with different amino acids; (w) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, and 84 are substituted with different amino acids; (x) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 94, and 96 are substituted with different amino acids; (y) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, and 92 are substituted with different amino acids; (z) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 61, 81, and 92 are substituted with different amino acids; (aa) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, and 92 are substituted with different amino acids; (ab) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, and 92 are substituted with different amino acids; (ac) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 81, 84, and 92 are substituted with different amino acids; (ad) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 20, 38, 42, 81, and 92 are substituted with different amino acids; (ae) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, and 92 are substituted with different amino acids; (af) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 81, and 92 are substituted with different amino acids; (ag) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 84, and 92 are substituted with different amino acids; (ah) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 81, 88, and 92 are substituted with different amino acids; (ai) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 85, 86, and 92 are substituted with different amino acids; (aj) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, and 92 are substituted with different amino acids; (ak) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 86, and 92 are substituted with different amino acids; (al) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, and 92 are substituted with different amino acids; (am) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 81, and 92 are substituted with different amino acids; (an) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 74, 80, 81, and 92 are substituted with different amino acids; (ao) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 84, and 92 are substituted with different amino acids; (ap) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 45, 80, 85, 86, and 92 are substituted with different amino acids; (aq) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ar) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (as) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (at) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 80, 81, 85, 86, and 92 are substituted with different amino acids; (au) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (av) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (aw) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ax) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 74, 80, 81, and 92 are substituted with different amino acids; (ay) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (az) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 69, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (ba) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 84, 85, 86, 91, and 92 are substituted with different amino acids; (bb) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, 86, 92, 94, and 96 are substituted with different amino acids; (bc) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 19, 22, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bd) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 38, 80, 81, 85, 86, and 92 are substituted with different amino acids; (be) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bf) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 18, 22, 68, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bg) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bh) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 45, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bi) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, and 92 are substituted with different amino acids; (bj) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 92, and 95 are substituted with different amino acids; (bk) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bl) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 43, 61, 80, 81, 85, 86, and 92 are substituted with different amino acids; (bm) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 38, 42, 80, 81, 85, 86, 91, 92, and 95 are substituted with different amino acids; (bn) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 35, 38, 42, 74, 80, 81, 82, 85, 86, and 92 are substituted with different amino acids; (bo) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 86, and 92 are substituted with different amino acids; or (bp) a method, wherein in native interleukin-2, the amino acid at position 1 is removed and the amino acids at positions 125, 80, 81, 85, and 86 are substituted with different amino acids, but the method is not limited thereto.

(91) The interleukin-2 analog and modification are the same as above.

(92) Still another aspect to implement the present invention provides an interleukin-2 analog which includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 3 to 106.

(93) The definitions of the interleukin-2 analog, modification, and analog represented by a SEQ ID NO are the same as above.

(94) Specifically, the interleukin-2 analog may include, essentially consist of, or consist of any one nucleotide sequence selected from the group consisting of SEQ ID NOS: 3 to 106, but the interleukin-2 analog is not limited thereto.

(95) Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence represented by General Formula 1 below:

(96) TABLE-US-00004 [General Formula 1] (General Formula 1, SEQ ID NO: 212) X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-X19-D-L- X22-M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42- X43-F-X45-M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K- P-L-E-X68-V-L-N-L-A-X74-S-K-N-F-H-X80-X81-P-R-X84- X85-X86-S-N-I-N-X91-X92-V-X94-E-X96-K-G-S-E-T-T-F- M-C-E-Y-A-D-E-T-A-T-I-V-E-F-L-N-R-W-I-T-F-S-Q-S-I- I-S-T-L-T

(97) wherein in General Formula 1 above, X1 is a deletion; X18 is leucine (L) or arginine (R); X19

is leucine (L) or tyrosine (Y); X22 is glutamic acid (E) or glutamine (Q); X38 is alanine (A), aspartic acid (D), or arginine (R); X42 is alanine (A), phenylalanine (F), lysine (K), or tryptophan (W); X43 is glutamic acid (E), lysine (K), or glutamine (Q); X45 is alanine (A) or tyrosine (Y); X61 is aspartic acid (D), glutamic acid (E), glutamine (Q), or arginine (R); X68 is aspartic acid (D) or glutamic acid (E); X74 is histidine (H) or glutamine (Q); X80 is phenylalanine (F), leucine (L), valine (V), or tyrosine (Y); X81 is aspartic acid (D), glutamic acid (E), or arginine (R); X84 is aspartic acid (D) or glutamic acid (E); X85 is alanine (A), glutamic acid (E), glycine (G), leucine (L), valine (V), tryptophan (W), or tyrosine (Y); X86 is alanine (A), glycine (G), isoleucine (I), or valine (V); X91 is threonine (T) or valine (V); X92 is phenylalanine (F), isoleucine (I), or tyrosine (Y); X94 is phenylalanine (F) or leucine (L); and X96 is phenylalanine (F) or leucine (L).
(98) Additionally, in General Formula 1 above, one or more amino acids may be added to threonine (T), which corresponds to X133, but the sequence is not limited thereto.
(99) Specifically, the interleukin-2 analog may include, essentially consist of, or consist of any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 15, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 56, 58, 59, 60, 62, 71, 72, 74, 75, 76, 77, 78, 85, 87, 89, 91, 92, 93, 94, 95, 98, 99, 100, 101, 103, 104, 105, and 106, but the interleukin-2 analog is not limited thereto.
(100) Such an interleukin-2 analog may have increased binding affinity for beta receptors compared to aldesleukin or native interleukin-2, but the binding affinity of the interleukin-2 analog is not limited thereto.
(101) In another embodiment, the interleukin-2 analog of the present invention may be: in General Formula 1 above, X43 is lysine (K); X45 is tyrosine (Y); X61 is aspartic acid (D), glutamic acid (E), or glutamine (Q); X68 is glutamic acid (E); X74 is glutamine (Q); X80 is phenylalanine (F) or leucine (L); X85 is leucine (L), valine (V), or tyrosine (Y); X86 is isoleucine (I) or valine (V); and X92 is phenylalanine (F) or isoleucine (I), but the interleukin-2 analog is not limited thereto.
(102) Specifically, the interleukin-2 analog is characterized in that it includes any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 10, 13, 14, 16, 17, 20, 21, 22, 32, 35, 36, 42, 53, 54, 87, 89, 91, 92, 93, 94, 98, 99, 100, 101, 103, 104, and 105.
(103) In the interleukin-2 analog of the present invention, one or more amino acids may be further added to a C-terminus thereof, but the interleukin-2 analog is not limited thereto.
(104) Still another aspect to implement the present invention provides an interleukin-2 analog which includes an amino acid sequence expressed by General Formula 2 below:
(105) TABLE-US-00005 [General Formula 2] (General Formula 2, SEQ ID NO: 213) X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-L-D-L-X22- M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-K-F-Y- M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-P-L-E-X68- V-L-N-L-A-Q-S-K-N-F-H-F-X81-P-R-D-X85-X86-S-N-I-N- V-F-V-L-E-L-K-G-S-E-T-T-F-M-C-E-Y-A-D-E-T-A-T-I-V- E-F-L-N-R-W-I-T-F-S-Q-S-I-I-S-T-L-T wherein in General Formula 2 above, X1 is a deletion, X18 is leucine (L) or arginine (R); X22 is glutamic acid (E) or glutamine (Q); X38 is alanine (A) or arginine (R); X42 is phenylalanine (F) or lysine (K); X61 is aspartic acid (D) or glutamic acid (E); X68 is aspartic acid (D) or glutamic acid (E); X81 is aspartic acid (D) or glutamic acid (E); X85 is leucine (L) or valine (V); and X86 is isoleucine (I) or valine (V).
(106) Specifically, the interleukin-2 analog may include any one sequence selected from the group consisting of amino acid sequences of SEQ ID NO: 22, 42, 53, 87, 105, and 106, but the sequence of the interleukin-2 analog is not limited thereto.
(107) Additionally, in General Formula 2 above, one or more amino acids may be added to threonine (T), which corresponds to X133, or alternatively, in the interleukin-2 analog, one or more amino acids may be further added to a C-terminus thereof, but the present invention is not limited thereto.
(108) Meanwhile, unless otherwise required by context in the present specification, expressions such as “include”, “including”, “containing”, etc. mean that they include a specified integer or

group of integers, but it should be understood that these expressions do not exclude other integers or a set of integers.
(109) Hereinafter, the present invention will be described in more detail through examples. These Examples are only for describing the present invention in more detail, and the scope of the present invention is not limited by these Examples.
Example 1: Preparation of Expression Vectors for Native Interleukin-2 and Interleukin-2 Analogs
(110) For the preparation of expression vectors for native interleukin-2 encoding 133 amino acids, an interleukin-2 that was synthesized based on the reported interleukin-2 sequence (NM_000586.3; SEQ ID NO: 1) was cloned into the pET-22b vector (Novagen). Additionally, a novel interleukin-2 analog was prepared in which an amino acid(s) of interleukin-2 were modified using the interleukin-2 as a template. The PCR conditions for the amplification of the interleukin-2 analog were 16 cycles of a process consisting of 95° C. for 30 seconds, 55° C. for 60 seconds, and 65° C. for 6.5 minutes. In order to confirm whether the amino acid(s) at the desired position had been correctly substituted, sequence analysis was performed on the mutagenesis product obtained under the conditions above. As a result, it was confirmed that the modifications shown in Table 1 below were found based on the native type at the desired mutation positions for each interleukin-2 analog. The thus-obtained expression vectors were named pET22b-interleukin-2 analogs 1 to 105.
(111) Table 1 below shows the altered sequences of amino acids and analog names for each. In order to prepare these interleukin-2 analogs, forward (F) and reverse (R) primers were synthesized, and then PCR was performed to amplify each analog gene.
(112) In Table 1 below, analog 1 represents aldesleukin and primer nos 1 to 203 correspond to SEQ ID NOS: 214 to 417, respectively.
(113) TABLE-US-00006 TABLE 1 Types of interleukin-2 analog, positions for modification, and altered sequences thereof Positions for Modification and Altered Sequences Analog Thereof Primer No. 1 desA1, C125S 197, 198, 201, 202 2 desA1, C125S, S87C 197, 198, 201, 202, 149, 150 3 desA1, C125S, K32C 197, 198, 201, 202, 19, 20 4 desA1, C125S, K35C 197, 198, 201, 202, 21, 22 5 desA1, C125S, K43C 197, 198, 201, 202, 47, 48 6 desA1, C125S, K48C 197, 198, 201, 202, 53, 54 7 desA1, C125S, K49C 197, 198, 201, 202, 55, 56 8 desA1, C125S, K76C 197, 198, 201, 202, 73, 74 9 desA1, C125S, R38A 197, 198, 201, 202, 25, 26 10 desA1, C125S, F42K 197, 198, 201, 202, 35, 36 11 desA1, C125S, F42A 197, 198, 201, 202, 33, 34 12 desA1, C125S, R38A, F42K 197, 198, 201, 202, 25, 26, 35, 36 13 desA1, C125S, R38A, F42A 197, 198, 201, 202, 25, 26, 33, 34 14 desA1, C125S, L19Y, R38A, F42K 197, 198, 201, 202, 13, 14, 25, 26, 35, 36 15 desA1, C125S, R38A, F42K, D84E 197, 198, 201, 202, 25, 26, 35, 36, 109, 110 16 desA1, C125S, R38A, F42K, N88Q 197, 198, 201, 202, 25, 26, 35, 36, 153, 154 17 desA1, C125S, R38A, F42K, V91T 197, 198, 201, 202, 25, 26, 35, 36, 165, 166 18 desA1, C125S, R38A, F42K, E61Q 197, 198, 201, 202, 25, 26, 35, 36, 59, 60 19 desA1, C125S, R38A, F42K, R81D, I92F 197, 198, 201, 202, 25, 26, 35, 36, 97, 98, 167, 168 20 desA1, C125S, R38A, F42K, L85V, I86V, I92F 197, 198, 201, 202, 25, 26, 35, 36, 139, 140, 167, 168 21 desA1, C125S, R38A, F42K, L80F, R81D, L85V, 197, 198, 201, 202, 25, I86V, I92F 26, 35, 36, 79, 80, 167, 168 22 desA1, C125S, L12V, R38A, F42K 197, 198, 201, 202, 3, 4, 25, 26, 35, 36 23 desA1, C125S, L12F, R38A, F42K 197, 198, 201, 202, 1, 2, 25, 26, 35, 36 24 desA1, C125S, L19V, R38A, F42K 197, 198, 201, 202, 11, 12, 25, 26, 35, 36 25 desA1, C125S, L19F, R38A, F42K 197, 198, 201, 202, 9, 10, 25, 26, 35, 36 26 desA1, C125S, R38A, F42K, I89F 197, 198, 201, 202, 25, 26, 35, 36, 157, 158 27 desA1, C125S, R38A, F42K, V91F 197, 198, 201, 202, 25, 26, 35, 36, 163, 164 28 desA1, C125S, R38A, F42K, L94V 197, 198, 201, 202, 25, 26, 35, 36 29 desA1, C125S, R38A, F42K, Q126T 197, 198, 201, 202, 25, 26, 35, 36, 199, 200 30 desA1, C125S, R38A, R81D, I92F 197, 198, 201, 202, 27, 28, 97, 98, 169, 170 31 desA1, C125S, R38A, D84E 197, 198, 201, 202, 27, 28, 109, 110 32 desA1, C125S, R38A, R81D, D84E, I92F 197, 198, 201, 202, 27, 28, 95, 96, 169, 170 33 desA1, C125S, R38A, L80F 197, 198, 201, 202, 27, 28, 77, 78 34 desA1, C125S, R38A, L80F, D84E 197, 198, 201, 202, 27, 28, 77, 78, 109, 110 35 desA1, C125S, R38A, L94F, L96F 197, 198, 201, 202, 27, 28, 189, 190 36 desA1,

C125S, R38A, L94F, L96V 197, 198, 201, 202, 27, 28, 193, 194 37 desA1, C125S, R38A, L94F, L96I 197, 198, 201, 202, 27, 28, 191, 192 38 desA1, C125S, R38A, F42K, R81D, I92F, L94F, 197, 198, 201, 202, 25, L96F 26, 35, 36, 97, 98, 175, 176 39 desA1, C125S, R38A, F42K, R81D, I92F, L94F, 197, 198, 201, 202, 25, L96V 26, 35, 36, 97, 98, 179, 180 40 desA1, C125S, R38A, F42K, R81D, I92F, L94F, L96I 197, 198, 201, 202, 25, 26, 35, 36, 97, 98, 177, 178 41 desA1, C125S, L80F, R81D, L85V, I86V, I92F 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41, 42, 79, 80, 167, 168 42 desA1, C125S, R38A, F42K, R81E, I92F 197, 198, 201, 202, 25, 26, 35, 36, 103, 104, 169, 170 43 desA1, C125S, R38A, F42K, R81D, I92L 197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 183, 184 44 desA1, C125S, R38A, F42K, R81D, D84V, I92F 197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 113, 114, 169, 170 45 desA1, C125S, R38A, F42K, R81D, D84F, I92F 197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 111, 112, 169, 170 46 desA1, C125S, D20V, R38A, F42K, R81D, I92F 197, 198, 201, 202, 17, 18, 25, 26, 35, 36, 99, 100, 169, 170 47 desA1, C125S, D20F, R38A, F42K, R81D, I92F 197, 198, 201, 202, 15, 16, 25, 26, 35, 36, 99, 100, 169, 170 48 desA1, C125S, R38A, F42K, R81D, N88V, I92F 197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 155, 156, 169, 170 49 desA1, C125S, R38A, F42K, R81D, N88F, I92F 197, 198, 201, 202, 25, 26, 35, 36, 99, 100, 151, 152, 169, 170 50 desA1, C125S, F42K, L80F, R81D, L85V, I86V, I92F 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 79, 80, 139, 140, 169, 170 51 desA1, C125S, E61Q, R81D, I92F 197, 198, 201, 202, 59, 60, 99, 100, 169, 170 52 desA1, C125S, R38A, F42K, L80F, R81D, I92F 197, 198, 201, 202, 25, 26, 35, 36, 79, 80, 169, 170 53 desA1, C125S, R38A, F42K, L80F, R81D, D84E, 197, 198, 201, 202, 25, I92F 26, 35, 36, 79, 80, 95, 96, 169, 170 54 desA1, C125S, R38A, F42K, Q74H, R81D, I92F 197, 198, 201, 202, 25, 26, 35, 36, 71, 72, 99, 100, 169, 170 55 desA1, C125S, R38A, F42K, Q74H, L80F, R81D, 197, 198, 201, 202, 25, I92F 26, 35, 36, 71, 72, 79, 80, 169, 170 56 desA1, C125S, R38A, F42K, Y45A, Q74H, R81D, 197, 198, 201, 202, 25, I92F 26, 35, 36, 49, 50, 71, 72, 99, 100, 169, 170 57 desA1, C125S, R38A, F42K, Y45A, Q74H, L80F, 197, 198, 201, 202, 25, R81D, I92F 26, 35, 36, 49, 50, 71, 72, 79, 80, 169, 170 58 desA1, C125S, R38A, F42K, L80F, R81D, L85A, 197, 198, 201, 202, 25, I86A, I92F 26, 35, 36, 79, 80, 115, 116, 167, 168 59 desA1, C125S, R38A, F42K, L80F, R81D, L85A, 197, 198, 201, 202, 25, I86A, I92Y 26, 35, 36, 79, 80, 115, 116, 187, 188 60 desA1, C125S, R38A, Y45A, L80Y, L85A, I86A, 197, 198, 201, 202, 27, I92Y 28, 35, 36, 41, 42, 51, 52, 93, 94, 115, 116, 187, 188 61 desA1, C125S, R38A, F42K, L80Y, R81D, L85G, 197, 198, 201, 202, 25, I86V, I92Y 26, 35, 36, 87, 88, 125, 126, 187, 188 62 desA1, C125S, R38A, L80W, R81E, L85G, I86A, 197, 198, 201, 202, 27, I92F 28, 35, 36, 41, 42, 85, 86, 123, 124, 171, 172 63 desA1, C125S, R38A, F42K, L80D, R81E, L85T, 197, 198, 201, 202, 25, I86G, I92F 26, 35, 36, 75, 76, 133, 134, 167, 168 64 desA1, C125S, R38A, F42K, L80Y, R81N, L85V, 197, 198, 201, 202, 25, I86V, I92F 26, 35, 36, 91, 92, 139, 140, 167, 168 65 desA1, C125S, R38A, F42K, L80Y, R81E, L85V, 197, 198, 201, 202, 25, I86V, I92F 26, 35, 36, 89, 90, 139, 140, 167, 168 66 desA1, C125S, R38A, F42K, L80F, R81E, L85F, 197, 198, 201, 202, 25, I86V, I92F 26, 35, 36, 81, 82, 121, 122, 167, 168 67 desA1, C125S, R38A, F42K, L80Y, R81D, L85F, 197, 198, 201, 202, 25, I86V, I92W, E95D 26, 35, 36, 87, 88, 121, 122, 185, 186, 195, 196 68 desA1, C125S, R38A, F42K, L80F, R81E, L85I, 197, 198, 201, 202, 25, I86V, V91E, I92F 26, 35, 36, 81, 82, 127, 128, 159, 160, 167, 168 69 desA1, C125S, R38A, F42K, L80Y, R81E, L85F, 197, 198, 201, 202, 25, I86L, V91E, I92W, E95D 26, 35, 36, 89, 90, 119, 120, 161, 162, 185, 186, 195, 196 70 desA1, C125S, R38A, F42K, L80Y, R81D, L85V, 197, 198, 201, 202, 25, I86V, I92F 26, 35, 36, 87, 88, 139, 140, 167, 168 71 desA1, C125S, R38A, F42K, L80F, R81E, L85V, 197, 198, 201, 202, 25, I86V, I92F 26, 35, 36, 81, 82, 139, 140, 167, 168 72 desA1, C125S, R38A, F42K, L80F, R81D, L85V, 197, 198, 201, 202, 25, I86G, I92F 26, 35, 36, 79, 80, 135, 136, 167, 168 73 desA1, C125S, R38A, F42K, L80F, R81D, L85W, 197, 198, 201, 202, 25, I86V, I92F 26, 35, 36, 79, 80, 141, 142, 167, 168 74 desA1, C125S, R38D, F42K, L80Y, R81D, L85V, 197, 198, 201, 202, 31, I86V, I92F 32, 35, 36, 87, 88, 139, 140, 167, 168 75 desA1, C125S, R38A, F42K, Y45A, L80F, R81E, 197, 198, 201, 202, 25, L85V, I86V, I92F 26, 35, 36, 51, 52, 81, 82, 139, 140, 167, 168 76 desA1, C125S, R38A, F42K, K43Q, E61R, L80F, 197, 198, 201, 202, 25, R81D, L85V, I86G, I92F 26, 39, 40, 61, 62, 79, 80, 135, 136, 167, 168 77 desA1, C125S, R38A, F42K, K43E, E61R, L80F, 197, 198, 201, 202, 25, R81D, L85W, I86V, I92F 26, 37, 38, 61, 62, 79, 80, 143, 144, 167, 168 78 desA1, C125S, K35E, R38A, F42K, L80F, R81E, 197, 198, 201, 202, 23, L85V, I86V, I92F 24, 25, 26, 35, 36, 81, 82, 139, 140, 167, 168 79 desA1, C125S, K35E, R38A, F42K, Q74H, L80F, 197, 198, 201, 202, 23, R81E, L85V, I86V, I92F 24, 25, 26, 35, 36, 71, 72, 81, 82, 139, 140, 167, 168 80 desA1, C125S, K35E, R38A, F42K, Q74H, L80F, 197, 198, 201, 202, 23, R81E, P82G, L85V, I86V, I92F 24, 25, 26, 35, 36, 71, 72, 81, 82, 105, 106, 139, 140, 167, 168 81 desA1, C125S, K35E, R38A, F42K, Q74H, L80F, 197, 198, 201, 202, 23, R81E, P82V, L85V, I86V, I92F 24, 25, 26, 35, 36, 71, 72, 81, 82, 107, 108, 139, 140, 167, 168 82 desA1, C125S, L18R, Q22E, L80F, R81D, L85E, 197, 198, 201, 202, 7, 8, I86V, I92F 25, 26, 29, 30, 35, 36, 43, 44, 79, 80, 117, 118, 167, 168 83 desA1, C125S, L18R, L19R, Q22E, L80F, R81D, 197, 198, 201, 202, 5, 6, L85E, I86V, I92F 25, 26, 29, 30, 35, 36, 43, 44, 79, 80, 117, 118, 167, 168 84 desA1, C125S, L18R, Q22E, L80V, R81D, L85E, 197, 198, 201, 202, 7, 8, I86V, I92F 25, 26, 29, 30, 35, 36, 43, 44, 83, 84, 117, 118, 167, 168 85 desA1, C125S, L80F, R81E, L85V, I86V, I92F 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 167, 168 86 desA1, C125S, L18R, Q22E, L80F, R81E, L85V, 197, 198, 201, 202, 7, 8, I86V, I92F 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 167, 168 87 desA1, C125S, L18R, L19R, Q22E, L80F, R81E, 197, 198, 201, 202, 5, 6, L85V, I86V, I92F 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 167, 168 88 desA1, C125S, E61D, L80F, R81E, L85V, I86V, I92F 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 57, 58, 81, 82, 139, 140, 167, 168 89 desA1, C125S, R38A, E68Q, L80F, R81E, L85V, 197, 198, 201, 202, 27, I86V, I92F 28, 35, 36, 43, 44, 65, 66, 81, 82, 139, 140, 167, 168 90 desA1, C125S, F42W, L80F, R81E, L85V, I86V, 197, 198, 201, 202, 25, I92F 26, 29, 30, 45, 46, 81, 82, 139, 140, 167, 168 91 desA1, C125S, E61Q, L80F, R81E, L85V, I86V, 197, 198, 201, 202, 25, I92F 26, 29, 30, 35, 36, 43, 44, 59, 60, 81, 82, 139, 140, 167, 168 92 desA1, C125S, L80F, R81E, L85V, I86V, V91T, I92F 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 147, 148, 167, 168 93 desA1, C125S, L80F, R81E, D84E, L85V, I86V, 197, 198, 201, 202, 25, V91T, I92F 26, 29, 30, 35, 36, 43, 44, 81, 82, 101, 102, 139, 140, 167, 168 94 desA1, C125S, L80F, R81E, L85V, I86V, I92F, 197, 198, 201, 202, 25, L94F, L96F 26, 29, 30, 35, 36, 43, 44, 81, 82, 139, 140, 173, 174 95 desA1, C125S, V69G, L80F, R81E, L85V, I86V, 197, 198, 201, 202, 25, I92F 26, 29, 30, 35, 36, 43, 44, 67, 68, 81, 82, 139, 140, 167, 168 96 desA1, C125S, V69G, Q74A, L80F, R81E, L85V, 197, 198, 201, 202, 25, I86V, I92F 26, 29, 30, 35, 36, 43, 44, 69, 70, 81, 82, 139, 140, 167, 168 97 desA1, C125S, R38A, L80F, R81D, I92F 197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 79, 80, 167, 168 98 desA1, C125S, R38A, L80F, R81E, L85V, I92F 197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 81, 82, 137, 138, 167, 168 99 desA1, C125S, R38A, L80F, R81E, I86V, I92F 197, 198, 201, 202, 27, 28, 35, 36, 41, 42, 81, 82, 131, 132, 167, 168 100 desA1, C125S, L80F, R81E, L85Y, I86V, I92F 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41, 42, 81, 82, 145, 146, 167, 168 101 desA1, C125S, L80F, R81E, I86A, I92F 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41, 42, 81, 82, 129, 130, 167, 168 102 desA1, C125S, L80F, R81E, L85V, I86V 197, 198, 201, 202, 25, 26, 29, 30, 35, 36, 41, 42, 81, 82, 139, 140, 181, 182 103 desA1, C125S, L18R, Q22E, R38A, L80F, R81E, 197, 198, 201, 202, 7, 8, L85V, I86V, I92F 27, 28, 35, 36, 41, 42, 81, 82, 139, 140, 167, 168 104 desA1, C125S, L18R, Q22E, E61D, L80F, R81E, 197, 198, 201, 202, 7, 8, L85V, I86V, I92F 25, 26, 29, 30, 35, 36, 41, 42, 57, 58, 81, 82, 139, 140, 167, 168 105 desA1, C125S, L18R, Q22E, E68D, L80F, R81E, 197, 198, 201, 202, 7, 8, L85V, I86V, I92F 25, 26, 29, 30, 35, 36, 41, 42, 63, 64, 81, 82, 139, 140, 167, 168

(114) In Table 1 above, desA1 represents a deletion of alanine, which is the first amino acid in interleukin-2.

(115) Table 2 below shows full-length protein sequences of interleukin-2 analogs. The letters shown in bold in Table 2 represent the positions for modification.

(116) TABLE-US-00007 TABLE 2 Sequences of interleukin-2 analogs SEQ ID SEQ ID NO of NO of Analog Protein Sequence Proteins Nucleotides 1 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 2 107 TFKFYMPKKA TELKHLQCLE

EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 2 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 3 108 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLI**C**NIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 3 PTSSSTKKT QLQLEHLLLD LQMILNGINN Y**C**NPKLTRML 4 109 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 4 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNP**C**LTRML 5 110 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 5 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 6 111 TF**C**FYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 6 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 7 112 TFKFYMP**C**KA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 7 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 8 113 TFKFYMPK**C**A TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 8 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 9 114 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQS**C**NFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 9 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 10 115 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 10 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 11 116 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 11 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 12 117 T**A**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 12 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 13 118 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 13 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 14 119 T**A**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 14 PTSSSTKKT QLQLEHLL**Y**D LQMILNGINN YKNPKLT**A**ML 15 120 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 15 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 16 121 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPR**E**LISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 16 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 17 122 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLIS**Q**IN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 17 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 18 123 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN **T**IVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 18 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 19 124 T**K**KFYMPKKA TELKHLQCLE **Q**ELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 19 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 20 125 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL **D**PRDLISNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 20 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 21 126 T**K**KFYMPKKA

TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 21 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 22 127 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F D**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 22 PTSSSTKKT QVQLEHLLLD LQMILNGINN YKNPKLT**A**ML 23 128 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 23 PTSSSTKKT Q**F**QLEHLLLD LQMILNGINN YKNPKLT**A**ML 24 129 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 24 PTSSSTKKT QLQLEHLL**V**D LQMILNGINN YKNPKLT**A**ML 25 130 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 25 PTSSSTKKT QLQLEHLL**F**D LQMILNGINN YKNPKLT**A**ML 26 131 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 26 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 27 132 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISN**F**N VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 27 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 28 133 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN **F**IVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 28 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 29 134 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIV**V**ELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 29 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 30 135 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 30 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 31 136 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL **D**PRDLISNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 31 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 32 137 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPR**E**LISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 32 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 33 138 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL **D**PR**E**LISNIN VFVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 33 PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLT**A**ML 34 139 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F** RPRDLISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 34 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 35 140 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F** RPR**E**LISNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 35 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 36 141 TFKFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIV**F**E**F**KGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 36 PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLT**A**ML 37 142 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIV**F**E**V**KGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 37 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 38 143 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL RPRDLISNIN VIV**F**E**I**KGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 38 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 39 144 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL **D**PRDLISNIN VFV**F**E**F**KGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 39 PTSSSTKKT QLQLEHLLLD

LQMILNGINN YKNPKLT**A**ML 40 145 T**K**KFYMPKKA TELKHLQCLE
EELKPLEEVL NLAQSKNFHL **D**PRDLISNIN V**F**V**F**E**V**KGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 40 PTSSSTKKT QLQLEHLLLD LQMILNGINN
YKNPKLT**A**ML 41 146 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFHL
DPRDLISNIN V**F**V**F**E**I**KGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 41
PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 42 147 TFKFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFH**F** **D**PRD**VV**SNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 42 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLT**A**ML 43 148 T**K**KFYMPKKA TELKHLQCLE
EELKPLEEVL NLAQSKNFHL **E**PRDLISNIN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 43 PTSSSTKKT QLQLEHLLLD LQMILNGINN
YKNPKLT**A**ML 44 149 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL
DPRDLISNIN V**L**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 44
PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 45 150 T**K**KFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFHL **D**PR**V**LISNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 45 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLT**A**ML 46 151 T**K**KFYMPKKA TELKHLQCLE
EELKPLEEVL NLAQSKNFHL **D**PR**F**LISNIN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 46 PTSSSTKKT QLQLEHLLL**V** LQMILNGINN
YKNPKLT**A**ML 47 152 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL
DPRDLISNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 47
PTSSSTKKT QLQLEHLLL**F** LQMILNGINN YKNPKLT**A**ML 48 153 T**K**KFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFHL **D**PRDLISNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 48 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLT**A**ML 49 154 T**K**KFYMPKKA TELKHLQCLE
EELKPLEEVL NLAQSKNFHL **D**PRDLIS**V**IN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 49 PTSSSTKKT QLQLEHLLLD LQMILNGINN
YKNPKLT**A**ML 50 155 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFHL
DPRDLIS**F**IN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 50
PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLTRML 51 156 T**K**KFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFH**F** **D**PRD**VV**SNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 51 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLTRML 52 157 TFKFYMPKKA TELKHLQCLE
QELKPLEEVL NLAQSKNFHL **D**PRDLISNIN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 52 PTSSSTKKT QLQLEHLLLD LQMILNGINN
YKNPKLT**A**ML 53 158 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F**
DPRDLISNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 53
PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 54 159 T**K**KFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFH**F** **D**PR**E**LISNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 54 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLT**A**ML 55 160 T**K**KFYMPKKA TELKHLQCLE
EELKPLEEVL NLA**H**SKNFHL **D**PRDLISNIN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 55 PTSSSTKKT QLQLEHLLLD LQMILNGINN
YKNPKLT**A**ML 56 161 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLA**H**SKNFH**F**
DPRDLISNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 56
PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLT**A**ML 57 162 T**K**KF**A**MPKKA
TELKHLQCLE EELKPLEEVL NLA**H**SKNFHL **D**PRDLISNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 57 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLT**A**ML 58 163 T**K**KF**A**MPKKA TELKHLOCLE
EELKPLEEVL NLA**H**SKNFH**F** **D**PRDLISNIN V**F**VLELKGSE TTFMCEYADE

TATIVEFLNR WITF**S**QSIIS TLT 58 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 59 164 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F D**PRD**AA**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 59 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 60 165 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F D**PRD**AA**SNIN V**Y**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 60 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 61 166 T**K**KF**A**MPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y** RPRD**AA**SNIN V**Y**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 61 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 62 167 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y D**PRD**GV**SNIN V**Y**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 62 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 63 168 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**W E**PRD**GA**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 63 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 64 169 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**D E**PRD**TG**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 64 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 65 170 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y N**PRD**VV**SNIN V**Y**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 65 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 66 171 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 66 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 67 172 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**FV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 67 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 68 173 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y D**PRD**FV**SNIN V**W**VL**D**LKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 68 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 69 174 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**IV**SNIN **EF**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 69 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 70 175 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y E**PRD**FL**SNIN **EW**VL**D**LKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 70 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 71 176 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y D**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 71 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 72 177 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 72 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 73 178 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F D**PRD**VG**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 73 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 74 179 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F D**PRD**WV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 74 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**D**ML 75 180 T**K**KFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**Y D**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 75 PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLT**A**ML 76 181 T**K**KF**A**MPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 76 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 77 182 T**KQ**FYMPKKA TELKHLQCLE **R**ELKPLEEVL NLAQSKNFH**F**

DPRD**VG**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 77
PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 78 183 T**KE**FYMPKKA
TELKHLQCLE **R**ELKPLEEVL NLAQSKNFH**F D**PRD**WV**SNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 78 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNP**E**LT**A**ML 79 184 T**K**KFYMPKKA TELKHLOCLE
EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 79 PTSSSTKKT QLQLEHLLLD LQMILNGINN
YKNP**E**LT**A**ML 80 185 T**K**KFYMPKKA TELKHLOCLE EELKPLEEVL NLA**H**SKNFH**F**
EPRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 80
PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNP**E**LT**A**ML 81 186 T**K**KFYMPKKA
TELKHLQCLE EELKPLEEVL NLA**H**SKNFH**F EG**RD**VV**SNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 81 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNP**E**LT**A**ML 82 187 T**K**KFYMPKKA TELKHLQCLE
EELKPLEEVL NLA**H**SKNFH**F EV**RD**VV**SNIN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 82 PTSSSTKKT QLQLEHL**R**LD L**E**MILNGINN
YKNPKLTRML 83 188 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F**
DPRD**EV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 83
PTSSSTKKT QLQLEHL**RR**D L**E**MILNGINN YKNPKLTRML 84 189 TFKFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFH**F D**PRD**EV**SNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 84 PTSSSTKKT QLQLEHL**R**LD
L**E**MILNGINN YKNPKLTRML 85 190 TFKFYMPKKA TELKHLQCLE EELKPLEEVL
NLAQSKNFH**V D**PRD**EV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR
WITF**S**QSIIS TLT 85 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 86
191 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN
V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 86 PTSSSTKKT
QLQLEHL**R**LD L**E**MILNGINN YKNPKLTRML 87 192 TFKFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 87 PTSSSTKKT QLQLEHL**RR**D
L**E**MILNGINN YKNPKLTRML 88 193 TFKFYMPKKA TELKHLQCLE EELKPLEEVL
NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR
WITF**S**QSIIS TLT 88 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 89
194 TFKFYMPKKA TELKHLQCLE **D**ELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN
V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 89 PTSSSTKKT
QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 90 195 TFKFYMPKKA
TELKHLQCLE EELKPLE**Q**VL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 90 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLTRML 91 196 T**W**KFYMPKKA TELKHLQCLE
EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE
TATIVEFLNR WITF**S**QSIIS TLT 91 PTSSSTKKT QLQLEHLLLD LQMILNGINN
YKNPKLTRML 92 197 TFKFYMPKKA TELKHLOCLE **Q**ELKPLEEVL NLAQSKNFH**F**
EPRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 92
PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 93 198 TFKFYMPKKA
TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN **TF**VLELKGSE
TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 93 PTSSSTKKT QLQLEHLLLD
LQMILNGINN YKNPKLTRML 94 199 TFKFYMPKKA TELKHLQCLE EELKPLEEVL
NLAQSKNFH**F E**PR**EVV**SNIN **TF**VLELKGSE TTFMCEYADE TATIVEFLNR
WITF**S**QSIIS TLT 94 PTSSSTKKT QLQLEHLLLD LOMILNGINN YKNPKLTRML 95
200 TFKFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN
V**F**V**F**E**F**KGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 95 PTSSSTKKT
QLQLEHLLLD LQMILNGINN YKNPKLTRML 96 201 TFKFYMPKKA

TELKHLQCLE EELKPLEE**G**L NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 96 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 97 202 TFKFYMPKKA TELKHLQCLE EELKPLEE**G**L NLA**A**SKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 97 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 98 203 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F D**PRDLISNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 98 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 99 204 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**V**ISNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 99 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLT**A**ML 100 205 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRDL**V**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 100 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 101 206 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**YV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 101 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 102 207 TFKFYMPKKA TELKHLOCLE EELKPLEEVL NLAQSKNFH**F E**PRDL**A**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 102 PTSSSTKKT QLQLEHLLLD LQMILNGINN YKNPKLTRML 103 208 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN VIVLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 103 PTSSSTKKT QLQLEHL**R**LD L**E**MILNGINN YKNPKLT**A**ML 104 209 TFKFYMPKKA TELKHLQCLE EELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 104 PTSSSTKKT QLQLEHL**R**LD L**E**MILNGINN YKNPKLTRML 105 210 TFKFYMPKKA TELKHLQCLE **D**ELKPLEEVL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT 105 PTSSSTKKT QLQLEHL**R**LD L**E**MILNGINN YKNPKLTRML 106 211 TFKFYMPKKA TELKHLQCLE EELKPLE**D**VL NLAQSKNFH**F E**PRD**VV**SNIN V**F**VLELKGSE TTFMCEYADE TATIVEFLNR WITF**S**QSIIS TLT

Example 2: Expression of Interleukin-2 Analogs

(117) A recombinant interleukin-2 analog under the control of T7 promoter was expressed using the expression vectors prepared in Example 1.

(118) An expression *E. coli* strain, *E. coli* BL21 DE3 (*E. coli* B F-dcm ompT hsdS(rB-mB-) gal A(DE3); Novagen), was transformed with each recombinant interleukin-2 analog expression vector. As for the transformation method, a method recommended by Novagen was used. Each single colony, in which each recombinant expression vector was transformed, was collected, inoculated into a 2X Luria Broth medium containing ampicillin (50 μg/mL), and cultured at 37° C. for 15 hours. Each recombinant strain culture solution and the 2× LB medium containing 30% glycerol were mixed at a 1:1 (v/v) ratio, and each 1 mL of the mixture was dispensed into a cryo-tube, and stored at −150° C. This was used as a cell stock for the production of a recombinant protein.

(119) For the expression of each recombinant interleukin-2 analog, one vial of each cell stock was dissolved, inoculated into 500 mL of 2× LB, and cultured with shaking at 37° C. for 14 to 16 hours. When the absorbance value at 600 nm reached 4.0 or higher, the culture was terminated, and this was used as a seed culture solution. The seed culture was inoculated into 1.6 L of a fermentation medium, and initial fermentation was started Using a 5 L fermentor (Bioflo-320, NBS, USA). Culture conditions were maintained at a pH of 6.70 using a temperature of 37° C., an air volume of 2.0 L/min (1 vvm), a stirring speed of 650 rpm, and 30% aqueous ammonia. As for the fermentation process, when nutrients in the culture medium were limited, fed-batch culture was performed by adding an additional medium (feeding solution). The growth of the strain was

observed by absorbance, and a final concentration of 500 μM IPTG was introduced at an absorbance value of 70 or higher. The culture was performed further until for about 23 to 25 hours after the introduction of IPTG, and after termination of the culture, and the recombinant strain was recovered using a centrifuge and stored at −80° C. until use.

Example 3: Extraction and Refolding of Interleukin-2 Analogs

(120) In order to convert the interleukin-2 analogs from the interleukin-2 analog expressing *E. coli* obtained in Example 2 in a soluble form, cells were disrupted and refolded. Cell pellets corresponding to 100 mL of the culture were suspended in 1-200 mL of a lysis buffer solution (20 mM Tris-HCl (pH 9.0), 1 mM EDTA (pH 9.0), 0.2 M NaCl, 0.5% Triton X-100), and the recombinant *E. coli* cells were disrupted at 15,000 psi using a microfluidizer. After centrifugation at 13,900 g for 30 minutes, the supernatant was discarded, and the pellet was washed with 400 mL of a first washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0)). After centrifugation under the same conditions as above, the supernatant was discarded, and the pellet was washed with 400 mL of a second washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0), 2% Triton X-100). After centrifugation under the same conditions as above, the supernatant was discarded, and the pellet was washed with 400 mL of a third washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0), 1% sodium deoxycholorate). After centrifugation under the same conditions as above, the supernatant was discarded, and the pellet was washed with 400 mL of a fourth washing buffer solution (50 mM Tris-HCl (pH 8.0), 5 mM EDTA (pH 9.0), 1 M NaCl). The resultant was subjected to centrifugation under the same conditions as above and washing, and *E. coli* inclusion bodies were obtained therefrom. The pellet of the washed inclusion bodies was resuspended in 400 mL of soluble/reducing buffer (6 M guanidine, 100 mM Tris (pH 8.0), 2 mM EDTA (pH 9.0), 50 mM DTT) and stirred at 50° C. for 30 minutes. To the soluble/reduced interleukin-2 analogs, 100 mL of distilled water was added to dilute the 6 M guanidine to 4.8 M guanidine, and then the resultant was centrifuged at 13,900 g for 30 minutes and the pellet discarded to obtain only the solution therein. To the diluted solution was additionally added 185.7 mL of distilled water, and the 4.8 M guanidine was diluted to 3.5 M guanidine, and the pH was adjusted to 5.0 using 100% acetic acid. The pH-adjusted solution was stirred at room temperature for one hour. The solution with precipitated impurities was centrifuged at 13,900 g for 30 minutes, and the supernatant was discarded and the pellet washed with a final washing buffer solution (3.5 M guanidine, 20 mM sodium acetate (pH 5.0), 5 mM DTT). The resultant was centrifuged under the same conditions as above to obtain a pellet. The washed interleukin-2 analogs were dissolved in 400 mL of a refolding buffer solution (6 mM guanidine, 100 mM Tris (pH 8.0), 0.1 mM CuCl.sub.2). The refolding process was performed by stirring the mixed solution at 4° C. for 15 to 24 hours.

Example 4: Size-Exclusion Column Chromatography

(121) The interleukin-2 analog refolding solution obtained in Example 3 was concentrated to less than 1 mL to be applied to a size-exclusion column for purification. The column was equilibrated with a buffer solution (2 M guanidine, 100 mM Tris (pH 8.0)) before introducing with the refolding solution and was eluted by flowing a buffer solution thereto after the introduction of the refolding solution. Since the eluted sample contained guanidine, it was replaced with a stabilized solution (10 mM sodium acetate (pH 4.5), 5% trehalose), and the purity was measured through RP-HPLC and peptide mapping analysis. The sample was used in the experiment when its measured purity reached 80% or higher.

Example 5: Evaluation of Binding Affinity of Interleukin-2 Analogs for Receptors

(122) In order to measure the binding affinity of the interleukin-2 analogs obtained in Example 4 for each of interleukin-2 alpha receptors and beta receptors, surface plasmon resonance measurement (BIACORE T200, GE Healthcare) was used. The binding affinity of the prepared analogs for the alpha receptors and beta receptors was measured, and the binding affinity of each of the prepared analogs was compared with that of interleukin-2 analog 01 (aldesleukin).

(123) First, an anti-human immunoglobulin antibody (Abcam, #ab97221) was immobilized to CM5 chips (GE Healthcare) by as much as about 5,000 RU (resonance unit) through amine coupling, and then, the immunoglobulin antibody was finally immobilized by allowing the interleukin-2 alpha receptors (SYMANSIS, #4102H) or interleukin-2 beta receptors (SYMANSIS, #4122H), to each of which a human immunoglobulin Fc region was bound, to bind to each immunoglobulin antibody using an antigen-antibody binding reaction. Thereafter, the recombinant interleukin-2 analog prepared above was diluted at various concentrations and was flowed onto the CM5 chips, to which the interleukin-2 receptors were finally immobilized, to measure the binding affinity of each interleukin-2 receptor. The measurement of binding affinity consisted of measurements of an association rate constant (k.sub.a) and a dissociation rate constant (k.sub.d), in which the binding rate was measured by flowing each interleukin-2 analog at a flow rate of 10 L/min for 3 minutes while the dissociation rate was measured from each interleukin-2 receptor by flowing only the experimental buffer for the same period of time and at the same flow rate. After the measurement was completed, the binding affinity for the receptors was evaluated according to the 1:1 binding fitting model in the Biaevaluation program.
relative binding affinity(*K*.sub.D)(%)=binding affinity of analog 01(aldesleukin)(*K*.sub.D)/binding affinity of analog(*K*.sub.D)×100
(124) In Table 3 below, "cannot be defined" indicates that the corresponding physical quantity cannot be defined for the corresponding receptor because no binding to the receptor was observed in the surface plasmon resonance measurement.
(125) TABLE-US-00008 TABLE 3 Relative binding affinity of interleukin-2 analogs for interleukin-2 alpha or beta receptors compared to analog 01 (aldesleukin) Interleukin-2 Receptor Test Material Relative Binding Affinity (%) Alpha Receptor analog 01 100.0 analog 09 74.5 analog 12 cannot be defined analog 13 1.1 analog 15 cannot be defined analog 16 0.2 analog 17 29.6 analog 19 cannot be defined analog 20 cannot be defined analog 21 cannot be defined analog 31 5.0 analog 34 9.4 analog 35 31.7 analog 41 121.3 analog 52 cannot be defined analog 53 cannot be defined analog 86 71.1 analog 88 101.5 analog 90 98.4 analog 91 7.9 analog 92 97.3 analog 93 92.8 analog 95 10.7 analog 96 14.9 analog 97 18.8 analog 98 7.7 analog 99 19.9 analog 100 29.1 analog 101 24.7 analog 102 151.4 analog 103 6.1 analog 104 122.4 analog 105 246.8 Beta Receptor analog 01 100.0 analog 09 337.4 analog 12 166.2 analog 13 148.6 analog 14 129.7 analog 15 98.1 analog 16 1261.8 analog 17 9.4 analog 18 35.3 analog 19 455.0 analog 20 156.5 analog 21 14,084.2 analog 24 37.9 analog 25 21.7 analog 31 235.7 analog 34 321.8 analog 35 232.7 analog 41 22,776.2 analog 52 3,821.1 analog 53 690.7 analog 55 3,025.7 analog 57 2,569.7 analog 58 7,771.2 analog 59 1,533.5 analog 61 1,039.1 analog 70 10,199.2 analog 71 17,083.8 analog 73 1,591.8 analog 74 8,153.4 analog 75 9,571.2 analog 76 1,040.4 analog 77 644.4 analog 84 710.7 analog 86 18,745.8 analog 88 13,856.6 analog 90 12,776.2 analog 91 7,361.9 analog 92 1,510.3 analog 93 696.8 analog 94 35.5 analog 95 17.1 analog 96 229.3 analog 97 3,019.4 analog 98 11,084.5 analog 99 1,509.1 analog 100 2,534.1 analog 101 113.1 analog 102 4,452.0 analog 103 13,100.0 analog 104 25,439.8 analog 105 26,837.8
(126) As explicitly shown in the test results (FIGS. **1** and **2** and Table 3), it was confirmed that the interleukin-2 analogs of the present invention had no binding affinity, increased/reduced binding affinity for interleukin-2 alpha receptors, etc., thus showing an altered binding affinity for interleukin-2 alpha receptors compared to native interleukin-2 or aldesleukin. In contrast, as for the interleukin-2 beta receptors, the interleukin-2 analogs of the present invention showed a stronger binding affinity of up to 100-fold compared to native interleukin-2 or aldesleukin. From the above results, it was confirmed that the amino acid sequence of the interleukin-2 analog has an effect on its binding to the interleukin-2 alpha or beta receptors. These results suggest that the binding affinity for interleukin-2 receptors can be altered by substituting an amino acid at a specific position.
(127) These experimental results suggest that the interleukin-2 analogs according to the present

invention have altered binding affinity for interleukin-2 alpha receptors and interleukin-2 beta receptors and thus can be used in the development of various drugs based on the same.
(128) From the foregoing, one of ordinary skill in the art to which the present invention pertains will be able to understand that the present invention may be embodied in other specific forms without modifying the technical concepts or essential characteristics of the present invention. In this regard, the exemplary embodiments disclosed herein are only for illustrative purposes and should not be construed as limiting the scope of the present invention. On the contrary, the present invention is intended to cover not only the exemplary embodiments but also various alternatives, modifications, equivalents, and other embodiments that may be included within the spirit and scope of the present invention as defined by the appended claims.

Claims

1. An interleukin-2 analog comprising an amino acid sequence represented by General Formula 1 below: [General Formula 1] X1-P-T-S-S-S-T-K-K-T-Q-L-Q-L-E-H-L-X18-X19-D-L-X22-M-I-L-N-G-I-N-N-Y-K-N-P-K-L-T-X38-M-L-T-X42-X43-F-X45-M-P-K-K-A-T-E-L-K-H-L-Q-C-L-E-X61-E-L-K-P-L-E-X68-V-L-N-L-A- X74-S-K-N-F-H-X80-X81-P-R-X84-X85-X86-S-N-I-N-X91-X92-V-X94-E-X96-K-G-S-E-T-T-F-M-C-E-Y-A-D-E-T-A-T-I-V-E-F-L-N-R-W-I-T-F-S-Q-S-I-I-S-T-L-T (General Formula 1, SEQ ID NO: 212) wherein in General Formula 1 above, X1 is a deletion; X18 is leucine (L) or arginine (R); X19 is leucine (L) or tyrosine (Y); X22 is glutamic acid (E) or glutamine (Q); X38 is alanine (A); X42 is alanine (A), phenylalanine (F), lysine (K), or tryptophan (W); X43 is glutamic acid (E), lysine (K), or glutamine (Q); X45 is alanine (A) or tyrosine (Y); X61 is aspartic acid (D), glutamic acid (E), glutamine (Q), or arginine (R); X68 is aspartic acid (D) or glutamic acid (E); X74 is histidine (H) or glutamine (Q); X80 is phenylalanine (F); X81 is aspartic acid (D), glutamic acid (E), or arginine (R); X84 is aspartic acid (D) or glutamic acid (E); X85 is alanine (A), glutamic acid (E), glycine (G), leucine (L), valine (V), tryptophan (W), or tyrosine (Y); X86 is alanine (A), glycine (G), isoleucine (I), or valine (V); X91 is threonine (T) or valine (V); X92 is phenylalanine (F); X94 is phenylalanine (F) or leucine (L); and X96 is phenylalanine (F) or leucine (L).
2. The interleukin-2 analog of claim 1, wherein the interleukin-2 analog comprises any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 54, 56, 58, 59, 72, 74, 76, 77, 78, 98, and 100.
3. The interleukin-2 analog of claim 1, wherein in General Formula 1 above, X43 is lysine (K); X45 is tyrosine (Y); X61 is aspartic acid (D), glutamic acid (E), or glutamine (Q); X68 is glutamic acid (E); X74 is glutamine (Q); X80 is phenylalanine (F); X85 is leucine (L), valine (V), or tyrosine (Y); X86 is isoleucine (I) or valine (V); and X92 is phenylalanine (F).
4. The interleukin-2 analog of claim 3, wherein the interleukin-2 analog comprises any one sequence selected from the group consisting of amino acid sequences of SEQ ID NOS: 54, 72, 98, and 100.
5. The interleukin-2 analog of claim 1, wherein the interleukin-2 analog further comprises one or more amino acids at the C-terminus thereof.